DevOrSysAdminScripts

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

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Current git SHA1: a5743e310279ed652270f4419d7b030b5ceaf0a8
Code lines: 10026 total lines, 9070 not empty lines, 956 empty lines.

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1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spell_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_listing.tex.tpl
│   ├── files_names_listing.txt
```

```
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   01234\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   012345
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   012345\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123456
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123456\
│   ├── a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │   an_see_that_it_works.txt
│   ├── funny.sh
│   ├── spampty_data.txt
│   └── yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│       that_you_can_see_that_it_works.txt
```

```
├── README.html
├── README.md
├── README.md.tpl
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 74 files

[ 12K Mar  9 16:37]  ./
├── [4.0K Mar  9 16:37]  build_and_checks_dependencies/
│   ├── [3.8K Mar  9 15:36]  build_and_checks.exec.sh.tpl*
│   ├── [3.8K Mar  9 16:23]  build_md_from_printable_md.exec.sh*
│   ├── [3.2K Mar  9 15:26]  check_shell_scripts_beginnings.libr.sh
│   ├── [5.0K Mar  5 21:47]  check_shell_scripts_indentation.libr.sh
│   ├── [3.4K Mar  5 21:17]  check_URLs.libr.sh
│   ├── [ 31K Mar  9 16:34]  common_build_and_checks.exec.sh*
│   ├── [ 25K Mar  9 16:08]  common_build_and_checks.exec.sh.tpl
│   ├── [9.1K Mar  8 22:03]  common_options.libr.sh
│   ├── [6.1K Mar  8 18:26]  comparisons.libr.sh
│   ├── [ 18K Mar  5 22:12]  create_PDF.exec.sh*
│   ├── [3.7K Mar  9 15:17]  generate_from_template.libr.sh
│   ├── [2.0K Mar  5 21:22]  get_common_text_glob_patterns.libr.sh
│   ├── [2.7K Mar  5 22:27]  grammar_and_spell_check.libr.sh
│   ├── [4.0K Mar  5 21:05]  licenses_templates/
│   │   ├── [ 13K Mar  5 21:05]  build_licenses_templates.exec.sh*
│   │   ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│   │   └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│   ├── [4.8K Mar  9 15:24]  lines_counts.libr.sh
│   ├── [ 13K Mar  9 15:21]  lines_filters.libr.sh
│   ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│   ├── [2.0K Mar  5 22:11]  overwrite_if_not_equal.libr.sh
│   ├── [2.3K Mar  8 21:54]  python_black_complement.libr.sh
│   ├── [2.8K Mar  9 15:29]  python_isort_complement.libr.sh
│   ├── [2.0K Mar  9 14:50]  repository_data.libr.sh
│   ├── [3.2K Mar  9 16:34]  shell_checks_complement.libr.sh
│   ├── [1.6K Feb  3 17:57]  split_score.exec.php
│   ├── [1.9K Feb  3 20:11]  split_score.libr.php
│   ├── [ 14K Mar  9 15:05]  strings_functions.libr.sh
│   ├── [3.8K Mar  5 21:28]  too_long_code_lines.libr.sh
│   ├── [4.2K Mar  9 16:05]  update_common_build_and_checks.exec.sh*
│   └── [5.5K Mar  9 14:39]  update_or_check_files_names_listing.exec.sh*
├── [4.0K Mar  9 16:34]  build_and_checks.exec.sh*
├── [4.0K Mar  9 16:37]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
```

```
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [ 16K Mar  8 17:00]  DevOrSysAdminScripts.html.tpl
│   ├── [ 32K Mar  9 16:37]  DevOrSysAdminScripts.tex
│   ├── [8.9K Mar  8 16:55]  DevOrSysAdminScripts.tex.tpl
│   ├── [ 14K Mar  9 16:09]  files_listing.tex.tpl
│   ├── [5.1K Mar  9 16:06]  files_names_listing.txt
│   ├── [  24 Jan 15 22:26]  .gitignore
│   ├── [5.2K Mar  8 16:52]  licenses_notes.txt
│   ├── [1.0K Mar  5 21:36]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.4K Mar  3 20:47]  repository_data.txt
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 167 Feb 19 18:39]  shellcheck.ini
│   └── [3.5K Mar  9 14:46]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [393K Mar  9 16:37]  DevOrSysAdminScripts.html
├── [352K Mar  9 16:37]  DevOrSysAdminScripts.pdf
├── [4.0K Mar  9 14:42]  git_helpers/
│   ├── [3.8K Mar  8 21:05]  git_checks.libr.sh
│   ├── [1.8K Feb 27 20:20]  post-commit*
│   ├── [ 14K Mar  9 14:42]  post_commit_renaming_comment.libr.sh
│   └── [1.4K Feb 19 18:42]  pre-commit*
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [4.0K Mar  9 15:15]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   0123\
│   ├── [   0 Mar  4 10:03]
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   01234\
│   ├── [   0 Mar  4 10:01]
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   012345
│   ├── [   0 Mar  4 10:37]
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   012345\
│   ├── [   0 Mar  4 10:37]
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   0123456
│   ├── [   0 Mar  4 10:37]
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   0123456\
│   ├── [   0 Mar  4 09:13]
```

```
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
│   ├── [1.3K Mar  9 15:15]  funny.sh
│   ├── [  94 Mar  3 19:56]  spampty_data.txt
│   └── [   0 Mar  4 09:13]
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── [8.0K Mar  9 15:34]  README.html
├── [4.2K Mar  9 15:34]  README.md
├── [4.2K Mar  9 15:34]  README.md.tpl
├── [9.1K Feb 26 15:05]  unicode.libr.php
├── [1.4K Mar  8 21:41]  upgrade_build_and_checks.exec.sh*
└── [1.5K Mar  8 20:25]  wget_sha512.libr.sh

6 directories, 74 files
```

2 Listing of files

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;

- The following dependencies are BSD-3-Clause licensed:
  phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
`https://www.gnu.org/software/findutils/`,
`https://savannah.gnu.org/projects/findutils/`.

grep (GNU grep) can be found at the following URLs:
`https://www.gnu.org/software/grep/`,
`https://savannah.gnu.org/projects/grep/`.

sed (GNU sed) can be found at the following URLs:
`https://www.gnu.org/software/sed/`,
`https://savannah.gnu.org/projects/sed/`.

ShellCheck can be found at the following URLs:
`https://www.shellcheck.net/`,
`https://github.com/koalaman/shellcheck`.

GNU Wget can be found at the following URLs:
`https://www.gnu.org/software/wget/`,
`https://savannah.gnu.org/projects/wget/`.

more of util-linux can be found at the following URLs:
`https://www.kernel.org/pub/linux/utils/util-linux/`,
`https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/`,
`https://github.com/util-linux/util-linux` official?

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.

If you see something that I missed regarding my dependencies, please tell me/email me :).

2.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
```

```
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    {
      printf "temp/\n\n"
    } > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
```

```
  @sha512_common_build_and_checks.exec.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed_point_build* ]]; then
  printf -- "--fixed_point_build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed_point_build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.2 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

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#
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
```

```
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "Searching md file: %s.\n" "${LFBFL_base_name}"
      fi
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed -Ez -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
  fi

  pandoc --from markdown --to html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    -o "${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed -i -E -e 's~(  <meta .*) />~\1>~g'\
```

```
            -e 's~<html .*>~<html lang="en">~'\
            -e '/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
             -e "${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

2.3 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
```

```
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # declare -r LFBFL_file_path=$(basename "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head -n 1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  local LFBFL_head
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
```

```
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    LFBFL_head=$(head -n 1 "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || printf "%s:File has wrong shell script beginning.\n"\
          "${LFBFL_file_path}"
  done
}
```

2.4 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
#!/usr/bin/env bash
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
```

```

check_one_shell_script_indentation(){
  declare -i LFBFL_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="workeduntilsomeasshole"
  # created a command named "workeduntilsomeasshole"
  # and called it without neither indentation, nor parameters
  # in a .sh script XD.
  # Manual justification of the lines L-4 and L-3 was a serendipity.
  # I just tried a joke and was lucky ;) :).
  local LFBFL_line_end
  local LFBFL_previous_line_end
  declare -i LFBFL_line_length
  declare -i LFBFL_line_end_length
  declare -i LFBFL_line_start_length
  local LFBFL_spacing
  declare -i LFBFL_offset
  local LFBFL_substring1
  local LFBFL_substring2
  declare -r LFBFL_pattern='^ *-e '
  # This grep would have been enough without some sed commands.
  declare -r LFBFL_s_lines=$(
    grep -EHn -B 1 '^(  )* ([^ ]|$)' "$1"
  )
  declare -a LFBFL_arr_lines
  mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
  readonly LFBFL_arr_lines

  for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
    if [[ "${LFBFL_some_line}" == "--" ]]; then
      # When switching between code fragments, we reinit.
      LFBFL_previous_line="workeduntilsomeasshole"
      continue
    fi
    # Note how it's funny that this loop always work because we check
    # shell scripts beginning in another script and thus we always
    # have a line before.
    if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]];
    then
      LFBFL_previous_line="${LFBFL_some_line}"
      continue
    fi
    # Remove the name of the file and then the line number
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
```

```
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_line_length=${#LFBFL_some_line}
    LFBFL_line_end_length=${#LFBFL_line_end}
    LFBFL_line_start_length=$((
      LFBFL_line_length-LFBFL_line_end_length
    ))
    LFBFL_offset=$((LFBFL_line_start_length-2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_offset+=1));;
      :) ((LFBFL_offset+=1));;
      [0-9]) ((LFBFL_offset+=2));;
      *) printf ":-(\n";;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf "%s\n" "${LFBFL_some_line}"
        LFBFL_file_with_error=1
      fi
    else
      printf "%s\n" "${LFBFL_some_line}"
      LFBFL_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
```

```
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

2.5 build_and_checks_dependencies/check_URLs.libr.sh

```
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```

```
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#
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# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )

  local LFBFL_pattern
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
```

```
      | xargs grep -H 'http:'\
      | grep -v "| xargs grep -H 'htt"\
      | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"
    # Last grep just above will remove false positives from
    # substitutions that fit on one line.

    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]] && printf "No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ -f "${LFBFL_file_path}" ]] || continue
      LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "Handling the file: %s.\n" "${LFBFL_file_path}"
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep -q "${LFBFL_substitution}" "${LFBFL_file_path}"; then
          sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            "${LFBFL_file_path}"
        fi
      done
    done
  done
}
```

2.6 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
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#
```

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#
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  declare -ar LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    # "$--work-directory={LFBFL_work_directory}" doesn't work
    "${LFBFL_work_directory_option}"
  )
```

```

source ./wget_sha512.libr.sh

LFBFL_subdir="build_and_checks_dependencies"
local LFBFL_file_name
local LFBFL_URL
local LFBFL_file_path

LFBFL_file_name="build_md_from_printable_md.exec.sh"
# LFBFL_script_download_URL
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='281f076b7e83ce4dff1f4c30bfa981d755d47c46b439a'
LFBFL_correct_sha512+='606344fd40d0d9337c6badbc41f6685ae1d543d94dd7'
LFBFL_correct_sha512+='a66d0c263abbf5bbed6dd1b703ff00350536ba8'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='e2d6a928cc8056440336e65416f2cb68aca5d67291446'
LFBFL_correct_sha512+='c134edcba52aca0400ef2f486330e02d40c6d4054042'
LFBFL_correct_sha512+='ad3c76444b0e261e0ef7bb4facd29dcc0bc7786'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='aec310eadf00f7f5349549d1feee8bebc6736b7a02c68'
LFBFL_correct_sha512+='e900a71743be58d5bfb8e73662cc7924501464c99e54'
LFBFL_correct_sha512+='c2f278bd643a6c8c83c2ce62a7f5eb0824880ae'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='35465ae8f66feab68ea43cb03372f6d14e152b1198e9f'
LFBFL_correct_sha512+='b0548f5c1aded656ba309980eaac5bbf4a17ecb3e500'
```

```
LFBFL_correct_sha512+='05ea66d9f464c45f3fd007c23aa4d3e7623e771'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='0f0bbd6f994c92ffb4cfc1748e2650857c88c3dc18b76'
LFBFL_correct_sha512+='7adb6a9962a6387ed470ddbfb8a6b981219659c1e19f'
LFBFL_correct_sha512+='2d0bbff7277251e08e8a165f4e98524ca7ba551'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='eb4f6855e09eec27832af9f375f5ea750f08bd04f7955'
LFBFL_correct_sha512+='e4a071216ca80772469ada184cfc96821d71f80f1b42'
LFBFL_correct_sha512+='01ea9e2da4279d07aaecb98739a9c59bbbb5b0a'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='b364c011469f2a308366f07c8e747ad51360fd2119c3d'
LFBFL_correct_sha512+='9addd2352aebbf8cfbaa4299485c237d34b879b48652'
LFBFL_correct_sha512+='c53b3854ce4746d24b2c7073b1986aa79e57e10'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='8846f6cfa5893288e3414e4e339e7881f62a1e13e1173'
LFBFL_correct_sha512+='1428a1847a3651fde29d71a998560d8109256c78b4c0'
LFBFL_correct_sha512+='b9e84786034479b53ba3493ca06ded078c683ee'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
```

```
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='fc041abf937df4c7cb1241c73e776e8bd07a5525bc9a9'
  LFBFL_correct_sha512+='29c83d597ddd6aa88a6e61a99d22ca88c11561f1cdbc'
  LFBFL_correct_sha512+='0fbf5df4794a2384f33e7e7dd5b65a53ab46f4a'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="grammar_and_spell_check.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='7238bffa4627414b6efb890950d3cfa125511124dc7a8'
  LFBFL_correct_sha512+='fc4c935c7dfd3da38ebfe8c3a06c65cf5b9275fe651e'
  LFBFL_correct_sha512+='1d98936c5bf9242b58b6679a05ecc37bef174b1'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  # /licenses_templates/ ---------------------------------------------
  declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
  declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
  LFBFL_file_name="build_licenses_templates.exec.sh"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='1e439c82e0232593038e973793ed01b63ace4d1913903'
  LFBFL_correct_sha512+='ceae9d112734f29d8f7c5b831d61f0ac1bf02be8bee9'
  LFBFL_correct_sha512+='1ee429522a8aca1f2b812c4f3eca66250ba41fe'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  LFBFL_file_name="license_file_header_AGPLv3+.tpl"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
  LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
  LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="license_file_header_GPLv3+.tpl"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
```

```
195    local LFBFL_correct_sha512
196    LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
197    LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
198    LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
199    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
200      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
201
202    LFBFL_file_name="license_file_header_LGPLv3+.tpl"
203    LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
204    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
205    local LFBFL_correct_sha512
206    LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
207    LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
208    LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
209    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
210      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
211    # ------------------------------------------------------------------
212
213    LFBFL_file_name="lines_counts.libr.sh"
214    LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
215    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
216    local LFBFL_correct_sha512
217    LFBFL_correct_sha512='adf26da440c9c258164bbbc157fd9eb327ecf7f449d0b'
218    LFBFL_correct_sha512+='4b3d7661f5a36295f47d17f8e3a7294a5e26af6b4f6f'
219    LFBFL_correct_sha512+='5302c40a33e584a584c2cb475f279d5d753f149'
220    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
221      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
222
223    LFBFL_file_name="lines_filters.libr.sh"
224    LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
225    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
226    local LFBFL_correct_sha512
227    LFBFL_correct_sha512='7ddf19d2df834ffa2246bc4e95b2b03263fa282c09221'
228    LFBFL_correct_sha512+='35f0db8c42461b4ee82cbdae2e76c7cd6328b22cbd33'
229    LFBFL_correct_sha512+='b773f34771baeb8212929a830560e33f80186de'
230    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
231      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
232
233    LFBFL_file_name="lines_maps.libr.sh"
234    LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
235    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
236    local LFBFL_correct_sha512
237    LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
238    LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
239    LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
240    wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
```

```
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="overwrite_if_not_equal.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='1f647d6da821c5ebc6e2cc1943d85ce8067eeec571079'
  LFBFL_correct_sha512+='59d37d9ee9d7d6f37be924ad278082b20284926ca5c4'
  LFBFL_correct_sha512+='d31441ca910a048267968e43548c9eb2a2729c2'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_black_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='afba2d9bff112d05e22dfbb7b4f50c9d925e4002473ce'
  LFBFL_correct_sha512+='df2f066368cac1dcf3943512cd39f9f07853b935c57f'
  LFBFL_correct_sha512+='acca56342e81700ec71e5076fb08ce65df62d10'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_isort_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='88d23cd7b9761c956c1bd33370d0457abf77c58f59c4a'
  LFBFL_correct_sha512+='c7c24f3ce6f2523a24d40e827df8e106cd27870242a6'
  LFBFL_correct_sha512+='327311471df4806231321020d8bb130084cf569'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="repository_data.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='793af27d8e658ed63148e6a3b45ba9b4eda1ddd3ede22'
  LFBFL_correct_sha512+='0b2af3c6a72e46d89b3565cef6e074cf1f9fcaccc7e4'
  LFBFL_correct_sha512+='dca98da471003bebeaabde704412ced55728094'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="shell_checks_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
```

```
LFBFL_correct_sha512='edf5b8042b6cf5fb0a2391b13aa863700543fd87e7cc7'
LFBFL_correct_sha512+='2971faf45dc29c75a1f18201c3a8dfcd9b30153a0300'
LFBFL_correct_sha512+='7a13741c898af552df66499ffddec99a87ee3ac'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.exec.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2f7ff107a43716109b1acbd7d89870d05577af00a981a'
LFBFL_correct_sha512+='05f162a054c96387fb99544f8225c333b1212f74326c'
LFBFL_correct_sha512+='f9ec6196e796f6257dfbba9347ac8c57b4f09db'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.libr.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='1413898a4fb4f31ca17993fd6110f7e1f32adf31c22c4'
LFBFL_correct_sha512+='7fa46a31d3c1511a3112a611e04fac19bb838d4476d9'
LFBFL_correct_sha512+='ecf77a28dd48d3d2732df7392c50e38262852d7'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2bcc71452545f83cbde9f528c8c638c8026c79f0cebcb'
LFBFL_correct_sha512+='959706b17db02b04fb3101349dd15814df130c192f55'
LFBFL_correct_sha512+='d475aef749bda6905c92da97a0deb1237dc6ae8'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='564069418a6f15fc95f08cb087269a480a4b67191523f'
LFBFL_correct_sha512+='78f74c8e1aabf9fd21a181a31948b9c042259575faf0'
LFBFL_correct_sha512+='9665a6582c72147a6de9948d8a2540311352840'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```
  LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='fd286f85e173d2cdc1099423e8e9978971346e237f7c4'
  LFBFL_correct_sha512+='75b076038d532a3a4a748df5e6e9f016ce08a11faadb'
  LFBFL_correct_sha512+='d86ca2653e831de43868d7c9c5818c7cfbca273'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  # shellcheck source=check_shell_scripts_beginnings.libr.sh
  source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
  # shellcheck source=check_shell_scripts_indentation.libr.sh
  source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
  # shellcheck source=check_URLs.libr.sh
  source "./${LFBFL_subdir}/check_URLs.libr.sh"
  # shellcheck source=common_options.libr.sh
  source "./${LFBFL_subdir}/common_options.libr.sh"
  # shellcheck source=comparisons.libr.sh
  source "./${LFBFL_subdir}/comparisons.libr.sh"
  # shellcheck source=generate_from_template.libr.sh
  source "./${LFBFL_subdir}/generate_from_template.libr.sh"
  # shellcheck source=get_common_text_glob_patterns.libr.sh
  source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
  # shellcheck source=grammar_and_spell_check.libr.sh
  source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
  # shellcheck source=lines_counts.libr.sh
  source "./${LFBFL_subdir}/lines_counts.libr.sh"
  # shellcheck source=lines_filters.libr.sh
  source "./${LFBFL_subdir}/lines_filters.libr.sh"
  # shellcheck source=lines_maps.libr.sh
  source "./${LFBFL_subdir}/lines_maps.libr.sh"
  # shellcheck source=overwrite_if_not_equal.libr.sh
  source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
  # shellcheck source=python_black_complement.libr.sh
  source "./${LFBFL_subdir}/python_black_complement.libr.sh"
  # shellcheck source=python_isort_complement.libr.sh
  source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
  # shellcheck source=shell_checks_complement.libr.sh
  source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
  # shellcheck source=strings_functions.libr.sh
  source "./${LFBFL_subdir}/strings_functions.libr.sh"
  # shellcheck source=too_long_code_lines.libr.sh
  source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

```

```
enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

enhanced_set_shell_option pipefail\
  && trap 'enhanced_unset_shell_option pipefail' RETURN

printf "Building license headers\n"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

printf "Building README.md\n"
"./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
  "--work-directory=${LFBFL_work_directory}"\
  "--base-name=README"\
  "${LFBFL_verbose}"

printf "Building other MarkDown files\n"
local LFBFL_some_directory
declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
local LFBFL_s_files_paths
LFBFL_s_files_paths=$(
  find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
  | relevant_find
)
declare -a LFBFL_arr_files_paths
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    printf "Found template %s.\n" "${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
    LFBFL_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
      "--work-directory=${LFBFL_some_directory}"\
      "--base-name=${LFBFL_file_name}"\
      "${LFBFL_verbose}"
  done
fi

declare -i LFBFL_will_popd
pushd_to_work_directory
```

```
  LFBFL_will_popd=$?
  can_continue_after_enhanced_pushd || return

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_upgrade_venvs_ts
  declare -i LFBFL_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      LFBFL_upgrade_venvs=1
    fi
  else
    LFBFL_upgrade_venvs=1
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_upgrade_venvs=0
      fi
    fi
```

```
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options[@]}"

  printf -- "---Python---\n"
  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
```

```
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_no_toml=1
```

```
LFBFL_s_files_paths=$(
  find . -type f -name "pyproject.toml"\
  | relevant_find
)
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
      printf "Running mypy\n"
      LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
      mypy "${LFBFL_directory_path}"
    fi
    LFBFL_no_toml=0
  done
fi
if [[ LFBFL_no_toml -eq 1 ]]; then
  printf "Running mypy\n"
  mypy .
fi
if [[ -n "${LFBFL_mypy_venv}" ]]; then
  deactivate
fi

printf "Running bandit\n"
LFBFL_bandit_venv=""
grep_variable "${LFBFL_data_file_name}" bandit_venv\
  --result-variable-prefix="LFBFL_"\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_bandit_venv}" ]]; then
  if command -v deactivate
  then
    deactivate
  fi
  # shellcheck disable=SC1090,SC1091
  source "${LFBFL_bandit_venv}/bin/activate"
fi
if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade bandit
fi
bandit --ini build_and_checks_variables/bandit.ini\
  -b build_and_checks_variables/bandit_baseline.json\
  -r .
# Saving new baseline in temp if necessary.
bandit --ini build_and_checks_variables/bandit.ini\
  -f json -o build_and_checks_variables/temp/bandit_baseline.json\
  -r .
```

```
if [[ -n "${LFBFL_bandit_venv}" ]]; then
  deactivate
fi

printf "Running pylint\n"
LFBFL_pylint_venv=""
grep_variable "${LFBFL_data_file_name}" pylint_venv\
  --result-variable-prefix="LFBFL_"\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_pylint_venv}" ]]; then
  if command -v deactivate
  then
    deactivate
  fi
  # shellcheck disable=SC1090,SC1091
  source "${LFBFL_pylint_venv}/bin/activate"
fi
if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade pylint
fi
pylint --rcfile build_and_checks_variables/pylintrc.toml\
  --recursive=y .
if [[ -n "${LFBFL_pylint_venv}" ]]; then
  deactivate
fi

printf "Running ruff\n"
LFBFL_ruff_venv=""
grep_variable "${LFBFL_data_file_name}" ruff_venv\
  --result-variable-prefix="LFBFL_"\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_ruff_venv}" ]]; then
  if command -v deactivate
  then
    deactivate
  fi
  # shellcheck disable=SC1090,SC1091
  source "${LFBFL_ruff_venv}/bin/activate"
fi
if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade ruff
fi
ruff check --config build_and_checks_variables/ruff.toml
if [[ -n "${LFBFL_ruff_venv}" ]]; then
  deactivate
fi
```

```
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHPMD\n"
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  local LFBFL_phpmd_baseline="build_and_checks_variables/"
  LFBFL_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_phpmd_baseline

  local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

  phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  # Saving new baseline in temp if necessary.
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file "${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
  diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
    --result-variable-prefix="LFBFL_"
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

```

```
  printf "Analyzing too long lines\n"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options[@]}" \
    --max-line-length=${LFBFL_max_line_length}

  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©"”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_will_popd -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
```

```
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    declare -a LFBFL_files_for_Nu2
    mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
    readonly LFBFL_files_for_Nu2
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose "${LFBFL_files_for_Nu2[@]}"
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.7 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
```

```
#
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# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
```

```
fi
readonly LFBFL_verbose
readonly LFBFL_i_verbose
declare -ar LFBFL_some_common_options=(
  "${LFBFL_verbose}"
  # "$--work-directory={LFBFL_work_directory}" doesn't work
  "${LFBFL_work_directory_option}"
)

source ./wget_sha512.libr.sh

LFBFL_subdir="build_and_checks_dependencies"
local LFBFL_file_name
local LFBFL_URL
local LFBFL_file_path

LFBFL_file_name="build_md_from_printable_md.exec.sh"
# LFBFL_script_download_URL
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_md_from_printable_md.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_beginnings.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_indentation.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_URLs.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

```

```
LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_common_options.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_comparisons.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_create_PDF.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_generate_from_template.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_get_common_text_glob_patterns.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="grammar_and_spell_check.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_grammar_and_spell_check.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

# /licenses_templates/ ---------------------------------------------
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
```

```
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_build_licenses_templates.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_AGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_GPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_maps.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_black_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_isort_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="repository_data.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_repository_data.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_shell_checks_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.exec.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_split_score.exec.php@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```
233
234     LFBFL_file_name="split_score.libr.php"
235     LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
236     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
237     @sha512_split_score.libr.php@
238     wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
239       "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
240
241     LFBFL_file_name="strings_functions.libr.sh"
242     LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
243     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
244     @sha512_strings_functions.libr.sh@
245     wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
246       "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
247
248     LFBFL_file_name="too_long_code_lines.libr.sh"
249     LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
250     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
251     @sha512_too_long_code_lines.libr.sh@
252     wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
253       "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
254
255     LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
256     LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
257     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
258     @sha512_update_or_check_files_names_listing.exec.sh@
259     wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
260       "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
261     chmod +x "./${LFBFL_file_path}"
262
263     # shellcheck source=check_shell_scripts_beginnings.libr.sh
264     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
265     # shellcheck source=check_shell_scripts_indentation.libr.sh
266     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
267     # shellcheck source=check_URLs.libr.sh
268     source "./${LFBFL_subdir}/check_URLs.libr.sh"
269     # shellcheck source=common_options.libr.sh
270     source "./${LFBFL_subdir}/common_options.libr.sh"
271     # shellcheck source=comparisons.libr.sh
272     source "./${LFBFL_subdir}/comparisons.libr.sh"
273     # shellcheck source=generate_from_template.libr.sh
274     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
275     # shellcheck source=get_common_text_glob_patterns.libr.sh
276     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
277     # shellcheck source=grammar_and_spell_check.libr.sh
278     source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
```

```
  # shellcheck source=lines_counts.libr.sh
  source "./${LFBFL_subdir}/lines_counts.libr.sh"
  # shellcheck source=lines_filters.libr.sh
  source "./${LFBFL_subdir}/lines_filters.libr.sh"
  # shellcheck source=lines_maps.libr.sh
  source "./${LFBFL_subdir}/lines_maps.libr.sh"
  # shellcheck source=overwrite_if_not_equal.libr.sh
  source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
  # shellcheck source=python_black_complement.libr.sh
  source "./${LFBFL_subdir}/python_black_complement.libr.sh"
  # shellcheck source=python_isort_complement.libr.sh
  source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
  # shellcheck source=shell_checks_complement.libr.sh
  source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
  # shellcheck source=strings_functions.libr.sh
  source "./${LFBFL_subdir}/strings_functions.libr.sh"
  # shellcheck source=too_long_code_lines.libr.sh
  source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

  enhanced_set_bash_option extglob
  # shellcheck source=repository_data.libr.sh
  source "./${LFBFL_subdir}/repository_data.libr.sh"
  # shellcheck disable=SC2154,SC2309
  [[ enhanced_set_bash_option_extglob_result -eq 0 ]]\
    && enhanced_unset_bash_option extglob

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  printf "Building license headers\n"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Building README.md\n"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  printf "Building other MarkDown files\n"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
```

```
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        "--work-directory=${LFBFL_some_directory}"\
        "--base-name=${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  declare -i LFBFL_will_popd
  pushd_to_work_directory
  LFBFL_will_popd=$?
  can_continue_after_enhanced_pushd || return

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_upgrade_venvs_ts
  declare -i LFBFL_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
```

```
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      LFBFL_upgrade_venvs=1
    fi
  else
    LFBFL_upgrade_venvs=1
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options[@]}"

  printf -- "---Python---\n"
  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
```

```
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options[@]}"

```

```
  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_no_toml=0
    done
  fi
  if [[ LFBFL_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix="LFBFL_"\
```

```
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini build_and_checks_variables/bandit.ini\
    -b build_and_checks_variables/bandit_baseline.json\
    -r .
  # Saving new baseline in temp if necessary.
  bandit --ini build_and_checks_variables/bandit.ini\
    -f json -o build_and_checks_variables/temp/bandit_baseline.json\
    -r .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile build_and_checks_variables/pylintrc.toml\
    --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
```

```
555     LFBFL_ruff_venv=""
556     grep_variable "${LFBFL_data_file_name}" ruff_venv\
557       --result-variable-prefix="LFBFL_"\
558       --replace-line-returns-by=""
559     if [[ -n "${LFBFL_ruff_venv}" ]]; then
560       if command -v deactivate
561       then
562         deactivate
563       fi
564       # shellcheck disable=SC1090,SC1091
565       source "${LFBFL_ruff_venv}/bin/activate"
566     fi
567     if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
568       pip install --upgrade ruff
569     fi
570     ruff check --config build_and_checks_variables/ruff.toml
571     if [[ -n "${LFBFL_ruff_venv}" ]]; then
572       deactivate
573     fi
574     printf -- "---Python end---\n"
575
576     printf -- "---PHP---\n"
577     printf "Running PHPMD\n"
578     if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
579       composer global require phpmd/phpmd
580     fi
581
582     local LFBFL_phpmd_baseline="build_and_checks_variables/"
583     LFBFL_phpmd_baseline+="phpmd_baseline.xml"
584     readonly LFBFL_phpmd_baseline
585
586     local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
587     LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
588     readonly LFBFL_temp_phpmd_baseline
589
590     local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
591     LFBFL_phpmd_rulesets+="design,naming,unusedcode"
592     readonly LFBFL_phpmd_rulesets
593
594     phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
595       . text "${LFBFL_phpmd_rulesets}"
596     # Saving new baseline in temp if necessary.
597     rm "${LFBFL_temp_phpmd_baseline}"
598     phpmd --color --generate-baseline\
599       --baseline-file "${LFBFL_temp_phpmd_baseline}"\
600       . text "${LFBFL_phpmd_rulesets}"
```

```
  sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
  diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
    --result-variable-prefix="LFBFL_"
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  printf "Analyzing too long lines\n"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options[@]}" \
    --max-line-length=${LFBFL_max_line_length}

  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_will_popd -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
```

```
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
```

```
    declare -a LFBFL_files_for_Nu2
    mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
    readonly LFBFL_files_for_Nu2
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose "${LFBFL_files_for_Nu2[@]}"
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.8 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
```

```
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd requested: %s\n" "$1"
  declare -ig enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated1\n"
  is_top_dirstack_directory "$1" && {
    enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  }
```

```
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated2\n"

  pushd "$1" || {
    enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3" "$1"
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed\n"
  # shellcheck disable=SC2248
  return ${enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_popd requested: %s\n" "$1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed\n"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ enhanced_pushd_result -eq 0\
  || enhanced_pushd_result -eq 110\
  || enhanced_pushd_result -eq 111\
  ]]
}
```

```

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    readonly LFBFL_i_popd_result
    printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
    LFBFL_i_verbose=1
  fi
  # shellcheck disable=SC2034
  readonly LFBFL_i_verbose
}


```

```
# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --work-directory=* ]]; then
      LFBFL_work_directory=${LFBFL_arg#--work-directory=}
      LFBFL_work_directory=${LFBFL_work_directory:-.}
      # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
      LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
      break
    fi
  done
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
}

popd_from_work_directory(){
  enhanced_popd "${LFBFL_work_directory}" 3
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
```

```
#   if you need to unset it in some function.
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  declare -r LFBFL_result_name="enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  declare -r LFBFL_result_name="enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set +o "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

```
enhanced_set_bash_option(){
  # $1=optname some bash option for shopt
  declare -r LFBFL_result_name="enhanced_set_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
  # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
  if shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -s "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  declare -r LFBFL_result_name="enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

2.9 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_first=1
  local LFBFL_current
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ LFBFL_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_arg}"
      LFBFL_first=0
      continue
    fi
    if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
      return 0
    fi
  done
  return 1
}
```

```
51
52 some_distinct(){
53   ! equal "$@"
54 }
55
56 all_distinct(){
57   # Returns 1 if all the arguments strings are distinct.
58   declare -ir LFBFL_ij_max=$#
59   declare -i LFBFL_i
60   declare -i LFBFL_j
61   for ((LFBFL_i=1; LFBFL_i<=LFBFL_ij_max; ++LFBFL_i)); do
62     for ((LFBFL_j=1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
63       [[ LFBFL_i -eq LFBFL_j ]] && continue
64       [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
65     done
66   done
67   return 1
68 }
69
70 all_distinct2(){
71   # Returns 1 if all the arguments strings are distinct.
72   local LFBFL_element
73   local LFBFL_previous_element
74   if [[ $# -le 1 ]]; then
75     return 1
76   fi
77   declare -r LFBFL_s_sorted_arguments=$(
78     printf "%s\n" "$@" \
79     | sort
80   )
81   declare -a LFBFL_arr_sorted_arguments
82   mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
83   declare -i LFBFL_i
84   declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
85   LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
86   for ((LFBFL_i=1; LFBFL_i<LFBFL_i_max; ++LFBFL_i)); do
87     LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
88     if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
89       return 0
90     fi
91     LFBFL_previous_element="${LFBFL_element}"
92   done
93   return 1
94 }
95
96 all_distinct3(){
```

```
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
    | uniq -d
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
```

```
  # $1=$sort command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1 -r"\
    | head -1
}

min(){
  # Returns the minimum string among the arguments.
  # Renvoie la chaîne la plus petite parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ min sort a b c
  # a
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1"\
    | head -1
}

is_substring(){
  # $1=$string
  # $2=$substring
  declare -r LFBFL_var_1=$(
    printf "%s" "$1"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
  )
  declare -r LFBFL_var_2=$(
    printf "%s" "$2"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
  )
  local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php -r "${LFBFL_var_3}"
```

```
  return $?
}

is_subfile(){
  # $1=$file_name
  # $2=$sub_file_name
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php -r "${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

2.10 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
```

```
27 # shellcheck source=common_options.libr.sh
28 source "./${LFBFL_subdir}/common_options.libr.sh"
29 # shellcheck source=generate_from_template.libr.sh
30 source "./${LFBFL_subdir}/generate_from_template.libr.sh"
31 # shellcheck source=lines_counts.libr.sh
32 source "./${LFBFL_subdir}/lines_counts.libr.sh"
33 # shellcheck source=lines_filters.libr.sh
34 source "./${LFBFL_subdir}/lines_filters.libr.sh"
35 # shellcheck source=lines_maps.libr.sh
36 source "./${LFBFL_subdir}/lines_maps.libr.sh"
37 # shellcheck source=overwrite_if_not_equal.libr.sh
38 source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
39 # shellcheck source=strings_functions.libr.sh
40 source "./${LFBFL_subdir}/strings_functions.libr.sh"
41
42 create_PDF(){
43   # Options:
44   #   --verbose
45   #   --work-directory=""
46   declare -i LFBFL_i_verbose=0
47   get_verbose_option "$@"
48   local LFBFL_work_directory=""
49   get_work_directory_option "$@"
50   pushd_to_work_directory\
51     && trap 'popd_from_work_directory' RETURN
52   can_continue_after_enhanced_pushd || return
53
54   enhanced_set_shell_option pipefail\
55     && trap 'enhanced_unset_shell_option pipefail' RETURN
56
57   local LFBFL_subdir2="build_and_checks_variables"
58   local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
59   local LFBFL_repository_name=""
60   grep_variable "${LFBFL_data_file_name}" repository_name\
61     --result-variable-prefix="LFBFL_"\
62     --replace-line-returns-by=""
63
64   local LFBFL_repository_web_url=""
65   grep_variable "${LFBFL_data_file_name}" repository_web_url\
66     --result-variable-prefix="LFBFL_"\
67     --replace-line-returns-by=""
68
69   local LFBFL_repository_git_url=""
70   grep_variable "${LFBFL_data_file_name}" repository_git_url\
71     --result-variable-prefix="LFBFL_"\
72     --replace-line-returns-by=""
```

```
  local LFBFL_abstract=""
  grep_variable "${LFBFL_data_file_name}" abstract\
    --result-variable-prefix="LFBFL_"

  local LFBFL_acknowledgments=""
  grep_variable "${LFBFL_data_file_name}" acknowledgments\
    --result-variable-prefix="LFBFL_"

  local LFBFL_author_email=""
  grep_variable "${LFBFL_data_file_name}" author_email\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_name}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  local LFBFL_author_website=""
  grep_variable "${LFBFL_data_file_name}" author_website\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -r LFBFL_current_date=$(date -I"date")

  declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)

  declare -ir LFBFL_number_of_commits=$(
    git shortlog\
    | space_starting_lines\
    | wc -l
  )

  local LFBFL_redirect="/dev/stdout"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_redirect="/dev/null"
  fi
  {
    code_lines_count_all "$@"
    code_lines_count_not_empty "$@"
    code_lines_count_empty "$@"
```

```
119     } &> "${LFBFL_redirect}"
120
121     local LFBFL_number_of_lines="${code_lines_count_all_result}"
122     LFBFL_number_of_lines+=" total lines,"
123     LFBFL_number_of_lines+=" ${code_lines_count_not_empty_result}"
124     LFBFL_number_of_lines+=" not empty lines,"
125     LFBFL_number_of_lines+=" ${code_lines_count_empty_result}"
126     LFBFL_number_of_lines+=" empty lines."
127     readonly LFBFL_number_of_lines
128
129     local LFBFL_temp_path
130     # LFBFL_temp_path=$(realpath "./${LFBFL_subdir2}/temp") for pushd
131     LFBFL_temp_path="./${LFBFL_subdir2}/temp"
132     readonly LFBFL_temp_path
133
134     # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
135     # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
136     # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
137     tree -a --gitignore\
138       -I "node_modules/"\
139       -I "__pycache__/"\
140       -I ".mypy_cache/"\
141       -I ".ruff_cache/"\
142       -I ".git/"\
143       | replace_non_ascii_spaces\
144       > "${LFBFL_temp_path}/current_tree_light.txt"
145
146     tree -a -DFh --gitignore\
147       -I "node_modules/"\
148       -I "__pycache__/"\
149       -I ".mypy_cache/"\
150       -I ".ruff_cache/"\
151       -I ".git/"\
152       | replace_non_ascii_spaces\
153       > "${LFBFL_temp_path}/current_tree.txt.temp"
154
155     local LFBFL_s_file_paths
156     LFBFL_s_file_paths=$(
157       sed -Ez 's/(.)\\\n/\1/Mg' "./${LFBFL_subdir2}/files_names_listing.txt"\
158       | grep -v '^//'
159     )
160     readonly LFBFL_s_file_paths
161     declare -a LFBFL_arr_file_paths
162     mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"
163     readonly LFBFL_arr_file_paths
164
```

```
  local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
  LFBFL_temp_files_listing+="files_listing.tex.tpl.temp"
  readonly LFBFL_temp_files_listing
  : > "${LFBFL_temp_files_listing}"
  local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
  LFBFL_temp_files_listing2+="files_listing.html.tpl.temp"
  readonly LFBFL_temp_files_listing2
  : > "${LFBFL_temp_files_listing2}"
  # HTML <li> elements, hence "lis".
  local LFBFL_temp_files_lis
  LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.tpl"
  readonly LFBFL_temp_files_lis
  : > "${LFBFL_temp_files_lis}"
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} /
  declare -r LFBFL_score_command="${get_split_score_result}"
  local LFBFL_score_command_properties="${get_split_score_result2}"
  readonly LFBFL_score_command_properties
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} ':'
  declare -r LFBFL_score_command2="${get_split_score_result}"
  local LFBFL_score_command_properties2="${get_split_score_result2}"
  readonly LFBFL_score_command_properties2
  declare -r LFBFL_suffix='%'
  declare -i LFBFL_i=0
  local LFBFL_cleaned_path1
  local LFBFL_cleaned_path2
  local LFBFL_new_lines
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_path
  # Remove line returns here to keep lines short.
  for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
    printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
    # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
    # LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
    LFBFL_cleaned_path2=$(
      printf "%s" "${LFBFL_file_path}"\
      | sed -e 's/\//:/g' -e 's/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
```

```
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        "${LFBFL_score_command_properties2}"
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    {
      printf "\\subsection{\n"
      printf "%s\n" "${LFBFL_new_lines}"
      printf "}\n"
      printf "\\label{\n"
      printf "%s\n" "${LFBFL_new_lines2}"
      printf "}\n"
      printf "\n"
      printf "\\VerbatimInput[numbers=left,xleftmargin=-5mm]{\n"
      printf "%s\n" "${LFBFL_new_lines3}"
      printf "}\n"
      printf "\n"
      printf "\n"
    } >> "${LFBFL_temp_files_listing}"

    # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
```

```
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      printf "<h3 id=\"subsection2.%s\">2.%s\n" "${LFBFL_i}" "${LFBFL_i}"
      printf "%s\n" "${LFBFL_new_lines}"
      printf "</h3>\n"
      printf "<pre class=\"numbered_lines\">\n"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n"
      printf "\n"
      printf "\n"
    } >> "${LFBFL_temp_files_listing2}"
    {
      printf "      <li><a href=\"#subsection2.%s\">\n" "${LFBFL_i}"
      printf "%s\n" "${LFBFL_new_lines}"
      printf "      </a></li>\n"
    } >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "./${LFBFL_subdir2}/files_listing.tex.tpl"\
    "${LFBFL_temp_files_listing}"
  sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
  overwrite_if_not_equal\
    "${LFBFL_temp_path}/files_listing.html.tpl"\
    "${LFBFL_temp_files_listing2}"

  # We verify if some lines are beyond max_line_length characters
  # in current_tree_light.txt and current_tree.txt.
  declare -ar LFBFL_trees=(
    "current_tree_light.txt"
    "current_tree.txt.temp"
  )
  local LFBFL_tree_path
  local LFBFL_line
  local LFBFL_prefix
  declare -i LFBFL_prefix_last_position
  local LFBFL_char
```

```
  local LFBFL_line_start
  declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
  local LFBFL_file_name
  local LFBFL_s_lines
  declare -a LFBFL_arr_lines
  declare -i LFBFL_keep_length
  for LFBFL_tree in "${LFBFL_trees[@]}"; do
    LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
    # shellcheck disable=SC2248
    LFBFL_s_lines=$(grep '.\{'${LFBFL_overlength}'\}' "${LFBFL_tree_path}")
    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
    for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
      # printf "LFBFL_line: %s\n" "${LFBFL_line}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E -e 's/(.*)──[^─]+$/\1/g'
      )
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_prefix_last_position}:1}"
      # printf "LFBFL_char: %s\n" "${LFBFL_char}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_prefix}"\
        | sed -E -e 's/[^ ]+$//g'
      )
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_file_name=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E 's|.* (([a-zA-Z0-9\._/-]+).)$|\1|g'
      )
      # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
      LFBFL_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
      LFBFL_line_start=${LFBFL_line:0:${LFBFL_keep_length}}
      LFBFL_line_start=$(
        printf "%s" "${LFBFL_line_start}"\
        | sed -e 's/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
        # shellcheck disable=SC2248
```

```
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
        "${LFBFL_tree_path}.some_line.temp"\
        ""\
        --quiet
    done
    rm -f "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  declare -r LFBFL_html_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.html"

  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]];
  then
    # If there is a template, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]];
  then
    # Otherwise if there is a tex file, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    printf "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
  fi

  # Same logic with repository HTML file.
  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]];
  then
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
```

```
  elif [[ -f "./${LFBFL_repository_name}.html" ]]; then
    cp "./${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    printf "Neither .html.tpl in ./%s/, nor .html in ./\n"\
      "${LFBFL_subdir2}"
  fi

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_abstract}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/abstract_temp"

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_acknowledgments}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/acknowledgments_temp"

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
  # Tex filling:
  if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_tex_path_start}.1"

    insert_file_at_token "${LFBFL_tex_path_start}.1" @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_tex_path_start}.2"

    insert_file_at_token "${LFBFL_tex_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_tex_path_start}.3"

    insert_file_at_token "${LFBFL_tex_path_start}.3"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
```

```
      "${LFBFL_tex_path_start}.4"

    insert_file_at_token "${LFBFL_tex_path_start}.4"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_tex_path_start}.5"

    insert_file_at_token "${LFBFL_tex_path_start}.5"\
      @files_listing_VerbatimInput@\
      "./${LFBFL_subdir2}/files_listing.tex.tpl"\
      "${LFBFL_tex_path_start}.6"

    overwrite_if_not_equal\
      "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.6" 1
  fi

  # HTML filling:
  declare -i LFBFL_HTML_updated=0
  if [[ -f "${LFBFL_html_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_html_path_start}.1"

    insert_file_at_token "${LFBFL_html_path_start}.1" @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_html_path_start}.2"

    insert_file_at_token "${LFBFL_html_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_html_path_start}.3"

    insert_file_at_token "${LFBFL_html_path_start}.3"\
      @files_lis@ "${LFBFL_temp_files_lis}"\
      "${LFBFL_html_path_start}.4"

    insert_file_at_token "${LFBFL_html_path_start}.4"\
```

```
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_html_path_start}.5"

    insert_file_at_token "${LFBFL_html_path_start}.5"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_html_path_start}.6"

    insert_file_at_token "${LFBFL_html_path_start}.6"\
      @files_listing_HTMLPreInput@\
      "${LFBFL_temp_path}/files_listing.html.tpl"\
      "${LFBFL_html_path_start}.7"

    overwrite_if_not_equal "./${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.7" 1
    LFBFL_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_HTML_updated -eq 0 ]]; then
    return
  fi

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"
    done
  else
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
        > /dev/null
    done
  fi

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
```

```
}

create_PDF "$@"
```

2.11 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
```

```
41    {
42      printf "%s\n" "$3"
43      cat "$1"
44      printf "%s\n" "$4"
45    } > "${LFBFL_temp}"
46    overwrite_if_not_equal "$2" "${LFBFL_temp}"
47  }
48
49  generate_from_template_with_line_comments(){
50    # $1=base_file_name
51    # $2=target_file_name
52    # $3=line_comment_prefix
53    # $4=optional_post_processing
54    # $5=file_prefix
55    local LFBFL_temp
56    LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
57    readonly LFBFL_temp
58    if [[ -n "$5" ]]; then
59      printf "%s\n" "$5"\
60        > "${LFBFL_temp}"
61    fi
62    sed -e "s/^/$3/g" "$1"\
63      >> "${LFBFL_temp}"
64    if [[ -n "$4" ]]; then
65      eval "$4"
66    fi
67    overwrite_if_not_equal "$2" "${LFBFL_temp}"
68  }
69
70  split_file_in_two(){
71    # $1=$file_name
72    # $2=$token assume that token is the only thing on his line.
73    # $3=$file_name_part1
74    # $4=$file_name_part2
75    # Other arguments are options for grep.
76    declare -a LFBFL_grep_options=()
77    declare -i LFBFL_i=0
78    local LFBFL_arg
79    for LFBFL_arg in "$@"; do
80      ((++LFBFL_i))
81      if [[ LFBFL_i -lt 5 ]]; then
82        continue
83      fi
84      LFBFL_grep_options+=("${LFBFL_arg}")
85    done
86    declare -ir LFBFL_line_number=$(
```

```
    grep -n "${LFBFL_grep_options[@]}" -- "$2" "$1"\
    | head --lines=1\
    | cut -f 1 -d ':'
  )
  declare -ir LFBFL_line_count=$(ll_wc -l -n "$1")
  declare -ir LFBFL_lines_after=$((
    LFBFL_line_count - LFBFL_line_number
  ))
  head --lines="$((LFBFL_line_number - 1))" "$1"\
    > "$3"
  tail --lines="${LFBFL_lines_after}" "$1"\
    > "$4"
}

insert_file_at_token(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_to_insert
  # $4=$file_name_result
  # Options:
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}" --fixed-strings
  cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}" "" "" "$5"
  fi
}
```

2.12 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

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# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
```

```
    '**/*.xml'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.xml'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

2.13 build_and_checks_dependencies/grammar_and_spell_check.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spell_check(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
```

```
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_grammar_or_spell_checker_command=""
  grep_variable "$1" grammar_or_spell_checker_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_or_spell_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    return
  fi
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      printf "No file found.\n"
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

2.14 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
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```

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#
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# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  # Source
```

```
54    local LFBFL_license_source_subdir
55    LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
56    readonly LFBFL_license_source_subdir
57    local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
58    LFBFL_source_license_prefix+="license_file_header_"
59    readonly LFBFL_source_license_prefix
60    # Target
61    local LFBFL_license_target_subdir
62    LFBFL_license_target_subdir="${LFBFL_work_directory}/"
63    LFBFL_license_target_subdir="build_and_checks_variables/temp/"
64    readonly LFBFL_license_target_subdir
65    local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
66    LFBFL_target_license_prefix+="license_file_header_"
67    readonly LFBFL_target_license_prefix
68
69    declare -ar LFBFL_licenses=(
70      "AGPLv3+"
71      "GPLv3+"
72      "LGPLv3+"
73    )
74
75    # ------------------------------------------------------------------
76    # Various language definitions for text file formats
77    declare -ar LFBFL_block_comment_languages=(
78      "c"
79      "py"
80    )
81    declare -ar LFBFL_block_comment_enters=(
82      '/*'
83      '"""'
84    )
85    declare -ar LFBFL_block_comment_exits=(
86      '*/'
87      '"""'
88    )
89    equal "${#LFBFL_block_comment_languages[@]}"\
90          "${#LFBFL_block_comment_enters[@]}"\
91          "${#LFBFL_block_comment_exits[@]}"
92    declare -i LFBFL_equal_result=$?
93    if [[ LFBFL_equal_result -eq 0 ]]; then
94      printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
95      return 1
96    fi
97
98    declare -ar LFBFL_line_comment_languages=(
99      "sh"
```

```
100       "tex"
101     )
102     declare -ar LFBFL_line_comment_prefixes=(
103       '# '
104       '% '
105     )
106     equal "${#LFBFL_line_comment_languages[@]}"\
107           "${#LFBFL_line_comment_prefixes[@]}"
108     LFBFL_equal_result=$?
109     if [[ LFBFL_equal_result -eq 0 ]]; then
110       printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
111       return 1
112     fi
113     # ------------------------------------------------------------------
114
115     # ------------------------------------------------------------------
116     # We generate the license header templates adapted to various text
117     # files formats.
118     # Kamoulox do endfor, Jacques Beauregard XD
119     # (C'est toujours mieux que l'inverse :) !)
120     # for do endfor -> for do done (Fort Doux Donne).
121     local LFBFL_some_license
122     local LFBFL_license_prefix
123     local LFBFL_license_prefix2
124     local LFBFL_extension
125     local LFBFL_license_file_name
126     local LFBFL_enter_string
127     local LFBFL_exit_string
128     local LFBFL_temp2
129     local LFBFL_prefix_string
130     local LFBFL_intermediate_file_name
131     local LFBFL_file_prefix
132     declare -i LFBFL_generate_from_template_result
133     for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
134       LFBFL_license_prefix="${LFBFL_source_license_prefix}"
135       LFBFL_license_prefix+="${LFBFL_some_license}"
136       LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
137       LFBFL_license_prefix2+="${LFBFL_some_license}"
138       for ((i=0; i<${#LFBFL_block_comment_languages[@]}; i++)); do
139         LFBFL_extension=${LFBFL_block_comment_languages[i]}
140         LFBFL_license_file_name="${LFBFL_license_prefix2}"
141         LFBFL_license_file_name+=".${LFBFL_extension}"
142         LFBFL_enter_string=${LFBFL_block_comment_enters[i]}
143         LFBFL_exit_string=${LFBFL_block_comment_exits[i]}
144         generate_from_template_with_block_comments\
145           "${LFBFL_license_prefix}.tpl"\
```

```
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_enter_string}" "${LFBFL_exit_string}"
      LFBFL_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_name}"
        elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_name}"
        fi
      fi
    done
    LFBFL_temp2=".generate_from_template_with_line_comments.temp"
    for ((i=0; i<${#LFBFL_line_comment_languages[@]}; i++)); do
      LFBFL_extension=${LFBFL_line_comment_languages[i]}
      LFBFL_license_file_name="${LFBFL_license_prefix2}"
      LFBFL_license_file_name+=".${LFBFL_extension}"
      LFBFL_prefix_string=${LFBFL_line_comment_prefixes[i]}
      LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
      LFBFL_intermediate_file_name+="${LFBFL_temp2}"
      LFBFL_file_prefix=""
      if [[ "${LFBFL_extension}" == "sh" ]]; then
        LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
      fi
      # The sed command is here to remove trailing spaces of previously
      # empty lines after comment prefix is added to the line.
      generate_from_template_with_line_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
        "${LFBFL_file_prefix}"
      LFBFL_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_name}"
        elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_name}"
        fi
      fi
    done
  done
  # ---------------------------------------------------------------------
```

```
192
193   # Now that base license templates generated languages license
194   # templates in working directory, we can go there.
195   pushd_to_work_directory\
196     && trap 'popd_from_work_directory' RETURN
197   can_continue_after_enhanced_pushd || return
198
199   local LFBFL_data_file_name="build_and_checks_variables/"
200   LFBFL_data_file_name+="repository_data.txt"
201   readonly LFBFL_data_file_name
202   local LFBFL_repository_name=""
203   grep_variable "${LFBFL_data_file_name}" repository_name\
204     --result-variable-prefix="LFBFL_"\
205     --replace-line-returns-by=""
206   local LFBFL_license=""
207   grep_variable "${LFBFL_data_file_name}" license\
208     --result-variable-prefix="LFBFL_"
209   local LFBFL_license2=""
210   grep_variable "${LFBFL_data_file_name}" license2\
211     --result-variable-prefix="LFBFL_"
212   local LFBFL_author_full_name=""
213   grep_variable "${LFBFL_data_file_name}" author_full_name\
214     --result-variable-prefix="LFBFL_"\
215     --replace-line-returns-by=" "
216
217   LFBFL_license_prefix="build_and_checks_variables/temp/"
218   LFBFL_license_prefix+="license_file_header_"
219   readonly LFBFL_license_prefix
220   LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
221   readonly LFBFL_license_prefix2
222   declare -r\
223     LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
224
225   # First year according to current state of git repository.
226   declare -ir LFBFL_first_year=$(
227     git log\
228     | grep 'Date:'\
229     | cut -f 8 -d ' '\
230     | tail -1
231   )
232   # Last year according to current state of git repository.
233   declare -ir LFBFL_last_year=$(
234     git log\
235     | grep 'Date:'\
236     | cut -f 8 -d ' '\
237     | head -1
```

```
)
local LFBFL_copyright_string
LFBFL_copyright_string="${LFBFL_first_year}-${LFBFL_last_year}"
LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
readonly LFBFL_copyright_string

prepare_filled_license_file_block(){
  # $1=license_file_name
  sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
      -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
      "$1.tpl"\
    > "$1.temp"
  overwrite_if_not_equal "$1" "$1.temp"
  declare -r LFBFL_result=$?
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    if [[ LFBFL_result -eq 1 ]]; then
      printf "License template %s updated.\n" "$1"
    elif [[ LFBFL_result -eq 2 ]]; then
      printf "License template %s created.\n" "$1"
    fi
  fi
  head --lines=-1 "$1"\
    | tail --lines=+2\
    > "$1.temp"
  # Strange world: tail --lines=-1 was working
  # to exclude first line.
  # I tested it months ago, and saw with my eyes the start and end
  # block comments removed with:
  # head --lines=-1 "$1" | tail --lines=-1.
  # Now, man page says to use --lines=+2 instead.
  # Welcome in the world where AI change your environment and your
  # documentation on the fly...
}

declare -Ar LFBFL_all_block_comment_languages=(
  ["c"]="c"
  ["php"]="c"
  ["py"]="py"
)

local LFBFL_key
local LFBFL_dest
local LFBFL_license_file_name2
declare -i LFBFL_not_subfile
declare -i LFBFL_not_subfile2
local LFBFL_file_path
```

```
284   local LFBFL_s_files_paths
285   declare -a LFBFL_arr_files_paths
286   for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
287     LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
288     LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
289     prepare_filled_license_file_block "${LFBFL_license_file_name}"
290     if [[ -n "${LFBFL_license2}" ]]; then
291       LFBFL_license_file_name2="${LFBFL_license_prefix3}"
292       LFBFL_license_file_name2+=".${LFBFL_dest}"
293       prepare_filled_license_file_block "${LFBFL_license_file_name2}"
294     fi
295
296     LFBFL_s_files_paths=$(
297       find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
298       | relevant_find
299     )
300     if [[ -z "${LFBFL_s_files_paths}" ]]; then
301       continue
302     fi
303     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
304     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
305       is_subfile "${LFBFL_file_path}"\
306         "${LFBFL_license_file_name}.temp"
307       LFBFL_not_subfile=$?
308       LFBFL_not_subfile2=1
309       if [[ -n "${LFBFL_license2}" ]]; then
310         is_subfile "${LFBFL_file_path}"\
311           "${LFBFL_license_file_name2}.temp"
312         LFBFL_not_subfile2=$?
313       fi
314       if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
315       then
316         printf "File %s has no/wrong license header?\n"\
317           "${LFBFL_file_path}"
318       fi
319     done
320     rm "${LFBFL_license_file_name}.temp"
321     if [[ -n "${LFBFL_license2}" ]]; then
322       rm "${LFBFL_license_file_name2}.temp"
323     fi
324   done
325
326   prepare_filled_license_file_line(){
327     # $1=LFBFL_license_file_name
328     sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
329         -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
```

```
330         "$1.tpl"\
331       > "$1.temp"
332     overwrite_if_not_equal "$1" "$1.temp"
333     declare -r LFBFL_result=$?
334     if [[ LFBFL_i_verbose -eq 1 ]]; then
335       if [[ LFBFL_result -eq 1 ]]; then
336         printf "License template %s updated." "$1"
337       elif [[ LFBFL_result -eq 2 ]]; then
338         printf "License template %s created." "$1"
339       fi
340     fi
341   }
342
343   declare -Ar LFBFL_all_line_comment_languages=(
344     ["sh"]="sh"
345     ["tex"]="tex"
346   )
347
348   for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
349     LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
350     LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
351     prepare_filled_license_file_line "${LFBFL_license_file_name}"
352     if [[ -n "${LFBFL_license2}" ]]; then
353       LFBFL_license_file_name2="${LFBFL_license_prefix3}"
354       LFBFL_license_file_name2+=".${LFBFL_dest}"
355       prepare_filled_license_file_line "${LFBFL_license_file_name2}"
356     fi
357
358     LFBFL_s_files_paths=$(
359       find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
360       | relevant_find
361     )
362     if [[ -z "${LFBFL_s_files_paths}" ]]; then
363       continue
364     fi
365     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
366     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
367       is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name}"
368       LFBFL_not_subfile=$?
369       LFBFL_not_subfile2=1
370       if [[ -n "${LFBFL_license2}" ]]; then
371         is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name2}"
372         LFBFL_not_subfile2=$?
373       fi
374       if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
375       then
```

```
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
  done
}

build_licenses_templates "$@"
```

2.15 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
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```

2.16 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

2.17 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

```
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```

2.18 build_and_checks_dependencies/lines_counts.libr.sh

```
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# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
```

```
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H -v 'a(?!a)a'
  done
}

all_self_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  # below and another lines, see SC2145 false positive
  all_code_lines "$@" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  all_self_code_lines "$@" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
```

```
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  all_self_code_lines "$@" \
    | not_empty_lines_after_file_name
}

code_lines_count_all(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi code_lines_count_all_result
  code_lines_count_all_result=$(
    all_self_code_lines\
    | wc -l
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${code_lines_count_all_result}
}

code_lines_count_empty(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi code_lines_count_empty_result
```

```
  # shellcheck disable=SC2119
  code_lines_count_empty_result=$(
    all_self_empty_code_lines\
    | wc -l
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${code_lines_count_empty_result}
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc -l
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${code_lines_count_not_empty_result}
}
```

2.19 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
```

```
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_number_only=1
    elif [[ ll_wc_var_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_number_only}"
  if [[ ll_wc_var_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed -e 's|domain|lists.gnu.org|' \
#       -e 's|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
```

```
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  # declare -ir LFBFL_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_lines_before}"
  # printf "%s\n" "${LFBFL_lines_after}"
  # if [[ LFBFL_lines_before -eq LFBFL_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent "${LFBFL_temp}" "${!#}"; then
    rm "${LFBFL_temp}"
    return
  fi
  mv "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=filepath
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below, you don't need to
  #   supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
```

```
150   declare -i LFBFL_enforce_line_starts=0
151   declare -i LFBFL_enforce_line_ends=0
152   declare -i LFBFL_quiet=0
153   local LFBFL_arg
154   for LFBFL_arg in "$@"; do
155     ((++LFBFL_i))
156     if [[ LFBFL_i -lt 3 ]]; then
157       continue
158     fi
159     if [[ "${LFBFL_arg}" == "--enforce-line-starts-with-fixed-string" ]];
160     then
161       LFBFL_enforce_line_starts=1
162       continue
163     fi
164     if [[ "${LFBFL_arg}" == "--enforce-line-ends-with-fixed-string" ]];
165     then
166       LFBFL_enforce_line_ends=1
167       continue
168     fi
169     if [[ "${LFBFL_arg}" == "-q" || "${LFBFL_arg}" == "--quiet" ]];
170     then
171       LFBFL_quiet=1
172       LFBFL_grep_options+=("${LFBFL_arg}")
173       continue
174     fi
175     LFBFL_grep_options2+=("${LFBFL_arg}")
176   done
177
178   if [[
179     LFBFL_enforce_line_starts -eq 0
180     && LFBFL_enforce_line_ends -eq 0
181     ]];
182   then
183     grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
184     return
185   fi
186   declare -r LFBFL_s_lines=$(
187     grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
188       "$2" "$1"
189   )
190   declare -a LFBFL_arr_lines
191   mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
192   readonly LFBFL_arr_lines
193
194   declare -i LFBFL_fixed_string_length=${#2}
195   declare -i LFBFL_i_result=1
```

```
196   local LFBFL_some_line
197   local LFBFL_line_part
198   for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
199     if [[ "${LFBFL_some_line}" == "$2" ]]; then
200       [[ LFBFL_quiet -eq 1 ]] || printf "%s\n" "${LFBFL_some_line}"
201       LFBFL_i_result=0
202       continue
203     fi
204     if [[
205       LFBFL_enforce_line_starts -eq 1
206       && LFBFL_enforce_line_ends -eq 1
207       ]];
208     then
209       continue
210     fi
211     if [[ LFBFL_enforce_line_starts -eq 1 ]]; then
212       LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_fixed_string_length}}
213     fi
214     if [[ LFBFL_enforce_line_ends -eq 1 ]]; then
215       LFBFL_line_part=${LFBFL_some_line: -${LFBFL_fixed_string_length}}
216     fi
217     if [[ "${LFBFL_line_part}" == "$2" ]]; then
218       [[ LFBFL_quiet -eq 1 ]] || printf "%s\n" "${LFBFL_some_line}"
219       LFBFL_i_result=0
220     fi
221   done
222   LFBFL_i_result=$((LFBFL_i_result | $?))
223   # shellcheck disable=SC2248
224   return ${LFBFL_i_result}
225 }
226
227 grep_variable(){
228   # $1=$file
229   # $2=$variable_name
230   # Options :
231   #   --replace-line-returns-by=""
232   #   --result-variable-prefix="LFBFL_" for example
233   declare -r LFBFL_regexp="(?<=^$2=).*$"
234   local LFBFL_arg
235   # printf "%s\n" "${LFBFL_regexp}"
236   local LFBFL_variable_value
237   if [[ "$*" == *--replace-line-returns-by=* ]]; then
238     local LFBFL_s_replace=""
239     for LFBFL_arg in "$@"; do
240       if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
241         LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
```

```
242         break
243       fi
244     done
245     LFBFL_variable_value=$(
246       grep -oP "${LFBFL_regexp}" "$1"\
247       | sed -Ez -e 's/\n/'"${LFBFL_s_replace}"'/Mg'
248     )
249     # We remove the trailing replace string,
250     # since the intent is only to replace the intermediate \n.
251     if [[ -n "${LFBFL_s_replace}" ]]; then
252     LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
253     fi
254   else
255     LFBFL_variable_value=$(grep -oP "${LFBFL_regexp}" "$1")
256   fi
257   local LFBFL_prefix=""
258   for LFBFL_arg in "$@"; do
259     if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
260       LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
261       break
262     fi
263   done
264   declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
265   # printf "%s : %s\n " "${LFBFL_variable_name}" "${LFBFL_variable_value}"
266   # The declare line may be dropped by a local or declare
267   # in outer scope.
268   # declare -g "${LFBFL_variable_name}"
269   # So the assignment must be done on another line.
270   printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
271   # printf "%s\n" "${!LFBFL_variable_name}"
272 }
273
274 empty_lines(){
275   grep '^$'
276 }
277
278 not_empty_lines(){
279   grep -v '^$'
280 }
281
282 space_starting_lines(){
283   grep '^ '
284 }
285
286 not_space_starting_lines(){
287   grep -v '^ '
```

```
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep -E '^( |$)'
}

not_spampty_lines(){
  grep '^[^ ]'
  # grep -v -E '^( |$)'
}

all_space_lines(){
  grep '^[ ]\+$'
}

not_all_space_lines(){
  grep -v '^[ ]\+$'
}

# All space or empty lines = aspampty
aspampty_lines(){
  grep '^[ ]*$'
}

not_aspampty_lines(){
  grep '[^ ]'
  # grep -v '^[ ]*$'
}

empty_lines_after_file_name(){
  grep '^[^:]\+:$'
}

not_empty_lines_after_file_name(){
  grep -v '^[^:]\+:$'
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep -v '^[^:]\+: '
}
```

```
spampty_lines_after_file_name(){
  grep -E '^[^:]+:( |$)'
}

not_spampty_lines_after_file_name(){
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep -v -E '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

not_all_space_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# :). Nice command :).

not_JS_dependencies_find(){
  grep -vE -e "(^|/)node_modules/"\
           -e "(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep -vE -e "^([^:]+/)?node_modules/"\
           -e "^([^:]+/)?package-lock\.json:"
}
```

```
not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
  not_JS_dependencies_grep
}

not_python_cache_find(){
  grep -vE -e "(^|/)__pycache__/"\
           -e "(^|/)\.mypy_cache/"\
           -e "(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
  grep -vE -e "^([^:]+/)?__pycache__/"\
           -e "^([^:]+/)?\.mypy_cache/"\
           -e "^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_python_cache_find
}

not_cache_grep(){
  not_python_cache_grep
}

not_git_find(){
  grep -vE "(^|/)\.git/"
}

not_git_grep(){
  grep -vE "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep -vE "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep -vE "^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep -vE -e "(^|/)COPYING$"\
```

```
          -e "(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep -vE -e "^([^:]+/)?COPYING:"\
           -e "^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.html$"
}

not_main_html_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.html:"
}

not_temp_file_find(){
  grep -vE "(^|/)build_and_checks_variables/temp/.*$"
```

```
}

not_temp_file_grep(){
  grep -vE "^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
}

relevant_find(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_find\
    | not_cache_find\
    | not_git_find\
    | not_archive_find\
    | not_temp_file_find
}

relevant_grep(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep
}
```

2.20 build_and_checks_dependencies/lines_maps.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
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```
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#
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
      -e 's/\xe2\x80\x80/ /g'\
      -e 's/\xe2\x80\x81/ /g'\
      -e 's/\xe2\x80\x82/ /g'\
      -e 's/\xe2\x80\x83/ /g'\
      -e 's/\xe2\x80\x84/ /g'\
      -e 's/\xe2\x80\x85/ /g'\
      -e 's/\xe2\x80\x86/ /g'\
      -e 's/\xe2\x80\x87/ /g'\
      -e 's/\xe2\x80\x88/ /g'\
      -e 's/\xe2\x80\x89/ /g'\
      -e 's/\xe2\x80\x8a/ /g'\
      -e 's/\xe2\x80\x8b/ /g'\
      -e 's/\xe2\x80\x8c/ /g'\
      -e 's/\xe2\x80\x8d/ /g'\
      -e 's/\xe2\x80\x8e/ /g'\
      -e 's/\xe2\x80\x8f/ /g'\
      -e 's/\xe2\x80\xa8/ /g'\
      -e 's/\xe2\x80\xa9/ /g'\
      -e 's/\xe2\x80\xaa/ /g'\
      -e 's/\xe2\x80\xab/ /g'\
      -e 's/\xe2\x80\xac/ /g'\
      -e 's/\xe2\x80\xad/ /g'\
      -e 's/\xe2\x80\xae/ /g'\
      -e 's/\xe2\x80\xaf/ /g'\
      -e 's/\xe2\x81\x9f/ /g'\
      -e 's/\xe2\x81\xa0/ /g'\
```

```
      -e 's/\xe2\x81\xa1/ /g'\
      -e 's/\xe2\x81\xa2/ /g'\
      -e 's/\xe2\x81\xa3/ /g'\
      -e 's/\xe2\x81\xa4/ /g'\
      -e 's/\xe2\x81\xa5/ /g'\
      -e 's/\xe2\x81\xa6/ /g'\
      -e 's/\xe2\x81\xa7/ /g'\
      -e 's/\xe2\x81\xa8/ /g'\
      -e 's/\xe2\x81\xa9/ /g'\
      -e 's/\xe2\x81\xaa/ /g'\
      -e 's/\xe2\x81\xab/ /g'\
      -e 's/\xe2\x81\xac/ /g'\
      -e 's/\xe2\x81\xad/ /g'\
      -e 's/\xe2\x81\xae/ /g'\
      -e 's/\xe2\x81\xaf/ /g'
}
```

2.21 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
```

```
28    # $1=$target_file_name
29    # $2=$temp_file_name
30    # $3=$keep_temp
31    # $4=$tree_mode : when the two files are the output of tree command,
32    # we ignore if all the differences are only on directories.
33    # $5=--quiet : we even don't want the message of diff saying that
34    #              files are distinct
35    # Return 0 target was the same
36    #        1 target was different
37    #        2 target did not already exist
38    if [[ ! -f "$1" ]]; then
39      if [[ -n "$3" ]]; then
40        cp -p "$2" "$1"
41      else
42        mv "$2" "$1"
43      fi
44      return 2
45    fi
46    declare -i LFBFL_is_equal
47    if [[ -n "$4" ]]; then
48      # Not setting pipefail since the result would still be incorrect.
49      # shellcheck disable=SC2312
50      diff "$1" "$2"\
51        | grep -E "^(>|<)"\
52        | grep -v "/$"
53      LFBFL_is_equal=${PIPESTATUS[2]}
54    else
55      if [[ -n "$5" ]]; then
56        diff -q "$1" "$2" > /dev/null
57      else
58        diff -q "$1" "$2"
59      fi
60      LFBFL_is_equal=1-$?
61    fi
62    if [[ LFBFL_is_equal -eq 1 ]]; then
63      if [[ -z "$3" ]]; then
64        rm "$2"
65      fi
66      return 0
67    fi
68    if [[ -n "$3" ]]; then
69      cp -p "$2" "$1"
70    else
71      mv "$2" "$1"
72    fi
73    return 1
```

```
}
```

2.22 build_and_checks_dependencies/python_black_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
```

```
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_bash_option globstar\
    && trap 'enhanced_unset_bash_option globstar' RETURN

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep -M --\
    $'def ([^"]|"(?!""))*"""([^"]|"(?!""))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  check_no_empty_line_after_python_function_docstring "$@"
}
```

2.23 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking import of _collections_abc is at the right place\n"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_temp=".check_collections_abc_place.temp"
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.py" -printf '%P\n'\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
```

```
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ -f "${LFBFL_file_path}" ]] || continue
    sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
      "${LFBFL_file_path}"\
      > "${LFBFL_file_path}${LFBFL_temp}"
    overwrite_if_not_equal "${LFBFL_file_path}"\
      "${LFBFL_file_path}${LFBFL_temp}"
  done
}

python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  check_collections_abc_place "$@"
}
```

2.24 build_and_checks_dependencies/repository_data.libr.sh

```
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```

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_name
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_bash_option extglob\
    && trap 'enhanced_unset_bash_option extglob' RETURN

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}
```

2.25 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_bash_option globstar\
    && trap 'enhanced_unset_bash_option globstar' RETURN

```

```
  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion."
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "%s\n" "${LFBFL_s}"

  pcre2grep -M -- $'\$\{#.*(\b|_)arr_(?!.*\[@\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  local LFBFL_arg
  declare -i LFBFL_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-line-length=}
      LFBFL_max_line_length=$((LFBFL_arg))
      break
    fi
  done
  declare -i LFBFL_max_previous_line_length=$((LFBFL_max_line_length - 5))

  enhanced_set_bash_option globstar\
    && trap 'enhanced_unset_bash_option globstar' RETURN

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
        "Checking that shell scripts do not contain misplaced 'then'.\n"

  local LFBFL_regexp
  LFBFL_regexp="^[^\n]{1,${LFBFL_max_previous_line_length}}"
  LFBFL_regexp+="(?<=\]\];)"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"

  pcre2grep -M -- "${LFBFL_regexp}" **/*.sh
```

```
}

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  check_no_size_of_array_first_element "$@"
  check_no_misplaced_then "$@"
}
```

2.26 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php
/*
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```

```
2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".
*/

require_once('./build_and_checks_dependencies/split_score.libr.php');

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// The first cast "(int) generate_split_score" is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1], (int) $argv[2], explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
```

2.27 build_and_checks_dependencies/split_score.libr.php

```
<?php
/*
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".
*/

function generate_split_score(
  $b_after_before,
  $i_max_length,
  $a_delimiter_strings_domain,
){
  return function(
    $s_delimiter_string,
    $i_cut_position,
    $b_is_cut_after,
  ) use ($b_after_before, $i_max_length, $a_delimiter_strings_domain){
    if(
      !in_array(
        $s_delimiter_string, $a_delimiter_strings_domain, true
      )
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_after_before){
      // When $b_after_before,
      // always way better to cut after '/' than before it.
      // When !$b_after_before (aka before_after),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
```

```
    return 1 + $i_cut_position;
  };
}
```

2.28 build_and_checks_dependencies/strings_functions.libr.sh

```
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# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"

split_line_at(){
  # $1=$line
  # $2=$position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
```

```
41    declare -g split_line_at_result_end="${1:$2}"
42    # printf "%s\n" "$split_line_at_result_beginning"
43    # printf "%s\n" "$split_line_at_result_end"
44  }
45
46  declare -gir SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM=1
47  declare -gir SPLIT_SCORE_PROPERTY_POSITION_UNIFORM=2
48  declare -gir SPLIT_SCORE_PROPERTY_LARGER_AFTER=4
49  declare -gir SPLIT_SCORE_PROPERTY_LARGER_BEFORE=8
50
51  split_score_simple(){
52    # $1=$after_before
53    # $2=$delimiters_strings_domain concatenated characters/delimiters
54    # $3=$delimiter_string a single character
55    # $4=$is_cut_after
56    declare -gi split_score_result
57    if [[ "$2" != *$3* ]]; then
58      split_score_result=0
59      return
60    fi
61    if [[ "$4" == "$1" ]]; then
62      split_score_result=2
63      return
64    fi
65    split_score_result=1
66  }
67
68  get_split_score_simple(){
69    # $1=$after_before
70    # $2=$max_length
71    # $3=$delimiters_strings_domain
72    declare -g get_split_score_result="split_score_simple $1 '$3'"
73    declare -gi get_split_score_result2
74    # get_split_score_result2=$((11-$1*4))
75    # if [[ $1 -eq 1 ]]; then
76    #   get_split_score_result2=7
77    # else
78    #   get_split_score_result2=11
79    # fi
80    get_split_score_result2=$((
81      SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM
82      + SPLIT_SCORE_PROPERTY_POSITION_UNIFORM
83      + SPLIT_SCORE_PROPERTY_LARGER_BEFORE
84      + (
85        $1 * (
86          SPLIT_SCORE_PROPERTY_LARGER_AFTER
```

```
87          - SPLIT_SCORE_PROPERTY_LARGER_BEFORE
88        )
89      )
90    ))
91  }
92
93  get_split_score_exec(){
94    declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
95    SPLIT_SCORE_EXEC+="/split_score.exec.php"
96  }
97
98  split_score(){
99    # $1=$after_before
100   # $2=$max_length
101   # $3=$delimiters_strings_domain concatenated characters/delimiters
102   # $4=$delimiter_string a single character
103   # $5=$cut_position
104   # $6=$is_cut_after
105   get_split_score_exec
106   declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
107   declare -gi split_score_result
108   split_score_result=$(eval "${LFBFL_command}")
109 }
110
111 # This function is currently not called,
112 # hence we need that shellcheck disable.
113 # shellcheck disable=SC2034
114 get_split_score(){
115   # $1=$after_before
116   # $1=$max_length
117   # $2=$delimiters_strings_domain
118   declare -g get_split_score_result="split_score $1 $2 '$3'"
119   declare -gi get_split_score_result2
120   # get_split_score_result2=$((9-$1*4))
121   get_split_score_result2=$((
122     SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM
123     + SPLIT_SCORE_PROPERTY_LARGER_BEFORE
124     + (
125       $1 * (
126         SPLIT_SCORE_PROPERTY_LARGER_AFTER
127         - SPLIT_SCORE_PROPERTY_LARGER_BEFORE
128       )
129     )
130   ))
131 }
132
```

```
133 split_line_at_most(){
134   # $1=$line
135   # $2=$max_length of beginning result string
136   # $3=$split_score_command
137   #   negative score means do not consider as a possible split
138   # $4=$split_score_command_properties :
139   #   These properties are flags.
140   #   If someday some property is not a flag,
141   #   then create split_score_command_properties2.
142   #   - null or 0 no property;
143   #   - 1 split score doesn't depend on which delimiter matches;
144   #   - 2 split score doesn't depend on the current position;
145   #   - 4 any split score after is always larger
146   #     than any split score before;
147   #   - 8 any split score before is always larger
148   #     than any split score after;
149   # $5=$forbidden_previous_character
150   # $6=$split_score_command_delimiter_strings_domain
151   #   optimisation if small domain
152   #   - this argument can be "null", then we will use $3 only instead,
153   #     in that case, only mono-character strings are considered,
154   #   - this argument can be an array of characters,
155   #     optimize accordingly,
156   #   - this argument can be an array of strings,
157   #     optimize accordingly,
158   #   - this argument can be an array of regexps,
159   #     optimize accordingly.
160   #     Not sure an array of regexps makes more sense than a single
161   #     one. Maybe they could be both cases with corresponding
162   #     optimizations. Same for a single character or substring.
163   #
164   # Note that we're always looking for the last position to split
165   # among maximum score;
166   # if some position gets as score from 2 delimiters,
167   # its score is always the maximum of the two corresponding scores.
168   declare -g split_line_at_most_result_start
169   declare -g split_line_at_most_result_end
170   declare -A LFBFL_positions
171   # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
172   # For my use case in bash scripts, I will need only an array of
173   # characters. See get_split_score().
174   if [[ -n "$6" ]]; then
175     printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
176   fi
177   if [[ $2 -ge ${#1} ]]; then
178     split_line_at_most_result_start=$1
```

```
179       split_line_at_most_result_end=""
180       return
181     fi
182     declare -r LFBFL_sort_command="sort --numeric-sort"
183     declare -ir LFBFL_length_minus_1=$((${#1} - 1))
184     declare -ir LFBFL_i_max=$(
185       min "${LFBFL_sort_command}" "${LFBFL_length_minus_1}" "$2"
186     )
187     declare -i LFBFL_i
188     declare -i LFBFL_j
189     local LFBFL_current_char
190
191     # At the beginning, each position has score 0.
192     for ((LFBFL_i=0; LFBFL_i<=LFBFL_i_max; ++LFBFL_i)) do
193       LFBFL_positions["${LFBFL_i}"]="0"
194     done
195     LFBFL_j=$((LFBFL_i+1))
196     LFBFL_positions["${LFBFL_j}"]="0"
197
198     # If we have a forbidden previous character,
199     # we mark the forbidden positions with -1.
200     if [[ -n "$5" ]]; then
201       for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
202         LFBFL_j=$((LFBFL_i+1))
203         LFBFL_current_char="${1:${LFBFL_i}:1}"
204         if [[ "${LFBFL_current_char}" == "$5" ]]; then
205           LFBFL_positions["${LFBFL_j}"]="-1"
206         fi
207       done
208     fi
209
210     if [[ "$4" == "7" ]]; then
211       for ((LFBFL_j=LFBFL_i_max; LFBFL_j>0; --LFBFL_j)) do
212         LFBFL_i=$((LFBFL_j-1))
213         LFBFL_current_char="${1:${LFBFL_i}:1}"
214         eval "$3 '${LFBFL_current_char}' 0"
215         # printf "%s|%s\n" "${split_score_result}" "${LFBFL_i}"
216         if [[ split_score_result -ge 1 ]]; then
217           # printf "%s|%s\n" "${split_score_result}" "${LFBFL_i}"
218           if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
219             LFBFL_positions["${LFBFL_i}"]=$(
220               max "${LFBFL_sort_command}"\
221                 "${LFBFL_positions["${LFBFL_i}"]}"\
222                 "${split_score_result}"
223             )
224           fi
```

```
225         fi
226         eval "$3 '${LFBFL_current_char}' 1"
227         # printf "%s|%s\n" "${split_score_result}" "${LFBFL_j}"
228         if [[ split_score_result -ge 1 ]]; then
229           # printf "%s|%s\n" "${split_score_result}" "${LFBFL_j}"
230           if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
231             LFBFL_positions["${LFBFL_j}"]=$(
232               max "${LFBFL_sort_command}"\
233                 "${LFBFL_positions["${LFBFL_j}"]}"\
234                 "${split_score_result}"
235             )
236             break
237           fi
238         fi
239       done
240     else
241       for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
242         LFBFL_j=$((LFBFL_i+1))
243         LFBFL_current_char="${1:${LFBFL_i}:1}"
244         eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
245         # printf "%s|%s\n" "${split_score_result}" "${LFBFL_i}"
246         if [[ split_score_result -ge 1 ]]; then
247           # printf "%s|%s\n" "${split_score_result}" "${LFBFL_i}"
248           if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
249             LFBFL_positions["${LFBFL_i}"]=$(
250               max "${LFBFL_sort_command}"\
251                 "${LFBFL_positions["${LFBFL_i}"]}"\
252                 "${split_score_result}"
253             )
254           fi
255         fi
256         eval "$3 '${LFBFL_current_char}' ${LFBFL_j} 1"
257         # printf "%s|%s\n" "${split_score_result}" "${LFBFL_j}"
258         if [[ split_score_result -ge 1 ]]; then
259           # printf "%s|%s\n" "${split_score_result}" "${LFBFL_j}"
260           if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
261             LFBFL_positions["${LFBFL_j}"]=$(
262               max "${LFBFL_sort_command}"\
263                 "${LFBFL_positions["${LFBFL_j}"]}"\
264                 "${split_score_result}"
265             )
266           fi
267         fi
268       done
269     fi
270     declare -ir LFBFL_max_score=$(
```

```
    max 'sort --numeric-sort' "${LFBFL_positions[@]}"
  )
  local LFBFL_value
  declare -i LFBFL_best_position
  for ((LFBFL_i=LFBFL_i_max; LFBFL_i>=0; --LFBFL_i)) do
    if [[ LFBFL_i -eq 0 ]]; then
      printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
    fi
    local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
    # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
    if [[ "${LFBFL_value}" == "${LFBFL_max_score}" ]]; then
      LFBFL_best_position=$((LFBFL_i))
      break
    fi
  done
  # shellcheck disable=SC2248
  split_line_at "$1" ${LFBFL_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # printf "%s\n" "$split_line_at_most_result_start"
  # printf "%s\n" "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=$new_lines
  # $2=$prefix
  # $3=$max_length
  # $4=$suffix
  # $5=$split_score_command
  # $6=$split_score_command_properties
  # $7=$forbidden_previous_character ('\' usually)
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
```

```
317     # This is logic, since it would be cumbersome to handle prefixing
318     # or not already splitted lines in this function
319     # (see repeated_split_last_line).
320     # The given max_length is the goal.
321     # This function adapts the effective max position for the split by
322     # taking into account the length of the suffix.
323     enhanced_set_shell_option pipefail\
324       && trap 'enhanced_unset_shell_option pipefail' RETURN
325
326     declare -g split_last_line_result="$1"
327     declare -ir LFBFL_max_length_plus=$(($3 + 1))
328     declare -ir LFBFL_length2=$(($3 - ${#4}))
329     declare -r LFBFL_regexp='.\{'"${LFBFL_max_length_plus}"'\}$'
330
331     # Testing if some line is too long.
332     printf "%s" "$1"\
333       | grep -q "${LFBFL_regexp}";
334     if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
335       return
336     fi
337
338     declare -r LFBFL_start=$(
339       printf "%s" "$1"\
340       | head --lines=-1
341     )
342     # printf "start: %s\n" "${LFBFL_start}"
343     declare -r LFBFL_last_line=$(
344       printf "%s" "$1"\
345       | tail --lines=1
346     )
347     # printf "last_line: %s\n" "${LFBFL_last_line}"
348     split_last_line_result=""
349     if [[ -n "${LFBFL_start}" ]]; then
350       split_last_line_result="${LFBFL_start}"
351       # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
352       # printf "%s %s\n" "${#split_last_line_result}"
353       #   "${split_last_line_result}"
354       split_last_line_result+=$'\n'
355       # something="${split_last_line_result}"
356       # printf "%s %s\n" "${#something}" "${something}"
357     fi
358     if [[ -n "$5" ]]; then
359       split_line_at_most "${LFBFL_last_line}" "${LFBFL_length2}"\
360         "$5" "$6" "$7"
361       split_last_line_result+="${split_line_at_most_result_start}"
362       split_last_line_result+="$4"
```

```
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_offset}}"
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

2.29 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
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```

```
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

too_long_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  local LFBFL_arg
  declare -i LFBFL_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-line-length=}
      LFBFL_max_line_length=$((LFBFL_arg))
      break
```

```
55      fi
56    done
57
58    enhanced_set_shell_option pipefail\
59      && trap 'enhanced_unset_shell_option pipefail' RETURN
60
61    get_COMMON_TEXT_FILES_GLOB_PATTERNS
62    local LFBFL_pattern
63    local LFBFL_long_line
64    local LFBFL_file_name
65    local LFBFL_line
66    local LFBFL_extension
67    local LFBFL_base_name
68    declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
69    local LFBFL_s_long_lines
70    declare -a LFBFL_arr_long_lines
71    for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
72      [[ LFBFL_i_verbose -eq 1 ]]\
73        && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
74      # shellcheck disable=SC2248
75      LFBFL_s_long_lines=$(
76        find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
77        | relevant_find\
78        | not_license_find\
79        | xargs grep -H '.\{'${LFBFL_overlength}'\}'
80      )
81      if [[ -z "${LFBFL_s_long_lines}" ]]; then
82        [[ LFBFL_i_verbose -eq 1 ]]\
83          && printf "No file found with long lines.\n"
84        continue
85      fi
86      mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
87      for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
88        LFBFL_file_name=${LFBFL_long_line%%:*} # Drop long ':*' suffix
89        LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
90        # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
91        LFBFL_extension=${LFBFL_file_name##*.} # Drop long '*.' prefix
92        LFBFL_base_name=${LFBFL_file_name%.*} # Drop short '.*' suffix
93        if [[ "${LFBFL_extension}" == "html" ]]; then
94          if [[ -f "${LFBFL_base_name}.md" ]]; then
95            continue
96          fi
97          if grep --quiet --fixed-strings -- "${LFBFL_line}"\
98            "./build_and_checks_variables/temp/files_listing.html.tpl"
99          then
100           continue
```

```
        fi
        printf "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if ! [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      else
        printf "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
}
```

2.30 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
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# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
```

```
32      printf "%s %s\n" "$0" "$*"
33      LFBFL_i_verbose=1
34    fi
35    readonly LFBFL_i_verbose
36
37    if [[ ! -o pipefail ]]; then
38      [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
39      set -o pipefail
40      trap 'set +o pipefail' RETURN
41    fi
42
43    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
44    cp "./${LFBFL_common_file_name}.tpl"\
45      "./${LFBFL_common_file_name}.temp"
46
47    declare -ar LFBFL_file_names=(
48      "build_md_from_printable_md.exec.sh"
49      "check_shell_scripts_beginnings.libr.sh"
50      "check_shell_scripts_indentation.libr.sh"
51      "check_URLs.libr.sh"
52      "common_options.libr.sh"
53      "comparisons.libr.sh"
54      "create_PDF.exec.sh"
55      "generate_from_template.libr.sh"
56      "get_common_text_glob_patterns.libr.sh"
57      "grammar_and_spell_check.libr.sh"
58      "licenses_templates/build_licenses_templates.exec.sh"
59      "licenses_templates/license_file_header_AGPLv3+.tpl"
60      "licenses_templates/license_file_header_GPLv3+.tpl"
61      "licenses_templates/license_file_header_LGPLv3+.tpl"
62      "lines_counts.libr.sh"
63      "lines_filters.libr.sh"
64      "lines_maps.libr.sh"
65      "update_or_check_files_names_listing.exec.sh"
66      "overwrite_if_not_equal.libr.sh"
67      "python_black_complement.libr.sh"
68      "python_isort_complement.libr.sh"
69      "repository_data.libr.sh"
70      "shell_checks_complement.libr.sh"
71      "split_score.exec.php"
72      "split_score.libr.php"
73      "strings_functions.libr.sh"
74      "too_long_code_lines.libr.sh"
75    )
76
77    local LFBFL_file_name
```

```
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_file_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
      | cut -f1 -d' '
    )
    LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    sed -i\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut -f1 -d' '
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}
```

```
update_common_build_and_checks "$@"
```

2.31 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
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# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
```

```
43    # Options:
44    #   --verbose
45    #   --work-directory=""
46    declare -i LFBFL_i_verbose=0
47    get_verbose_option "$@"
48    local LFBFL_work_directory=""
49    get_work_directory_option "$@"
50    pushd_to_work_directory\
51      && trap 'popd_from_work_directory' RETURN
52    can_continue_after_enhanced_pushd || return
53
54    declare -i LFBFL_write=0
55    if [[ "$*" == *--write* ]]; then
56      LFBFL_write=1
57    fi
58    readonly LFBFL_write
59
60    declare -i LFBFL_append=0
61    if [[ "$*" == *--append* ]]; then
62      LFBFL_append=1
63    fi
64    readonly LFBFL_append
65
66    enhanced_set_shell_option pipefail\
67      && trap 'enhanced_unset_shell_option pipefail' RETURN
68
69    LFBFL_subdir2="build_and_checks_variables"
70    local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
71    readonly LFBFL_listing
72    local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
73    readonly LFBFL_data_file_name
74
75    declare -i LFBFL_max_line_length
76    grep_variable "${LFBFL_data_file_name}" max_line_length\
77      --result-variable-prefix="LFBFL_"
78
79    if [[ LFBFL_write -eq 1 ]]; then
80      : > "${LFBFL_listing}"
81    fi
82    # Remove line returns that are here to keep lines short.
83    sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_listing}"\
84      > "${LFBFL_listing}.temp"
85    # shopt -s dotglob was needed at some point but I don't see why now.
86    # shellcheck disable=SC2248
87    get_split_score_simple 1 ${LFBFL_max_line_length} /
88    declare -r LFBFL_split_score_command="${get_split_score_result}"
```

```
89    local LFBFL_split_score_command_properties
90    LFBFL_split_score_command_properties="${get_split_score_result2}"
91    readonly LFBFL_split_score_command_properties
92    local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
93    readonly LFBFL_suffix
94    # There is always a $'\n' final suffix from >>.
95    # But when origin string ends with the suffix we must add something.
96
97    local LFBFL_s_file_paths
98    LFBFL_s_file_paths=$(
99      find . -type f -printf '%P\n'\
100     | relevant_find\
101     | sort
102   )
103   declare -a LFBFL_arr_file_paths
104   mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"
105
106   declare -r LFBFL_final_suffix=$'\n\\'
107   local LFBFL_file_path
108   local LFBFL_base_file_name
109   for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
110     git check-ignore -q "${LFBFL_file_path}" && continue
111     LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
112     [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
113       || continue
114     if [[ "${LFBFL_base_file_name}" == *.md ]]; then
115       if [[ -f "${LFBFL_file_path}.tpl" ]]; then
116         continue
117       fi
118     fi
119     if [[ LFBFL_i_verbose -eq 1 ]]; then
120       printf "Non-ignored file: %s\n" "${LFBFL_file_path}"
121     fi
122     # shellcheck disable=SC2248
123     repeated_split_last_line "${LFBFL_file_path}"\
124       ""\
125       ${LFBFL_max_line_length}\
126       "${LFBFL_suffix}"\
127       "${LFBFL_split_score_command}"\
128       "${LFBFL_split_score_command_properties}"\
129       ""\
130       "${LFBFL_final_suffix}"
131     if [[ LFBFL_write -eq 1 ]]; then
132       # shellcheck disable=SC2001
133       printf "%s\n" "${repeated_split_last_line_result}"\
134         >> "${LFBFL_listing}"
```

```
      continue
    fi
    if grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
        "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
    then
      :
    else
      printf "The file %s is not listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
      if [[ LFBFL_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        printf "%s\n" "${repeated_split_last_line_result}"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  LFBFL_s_file_paths=$(cat "${LFBFL_listing}.temp")
  mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"

  for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
    [[ "${LFBFL_file_path}" != '//'* ]] || continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    if ! [[ -f "${LFBFL_file_path}" ]]; then
      printf "The non-file %s is listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
    fi
  done

  rm "${LFBFL_listing}.temp"

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    cat "${LFBFL_listing}"
  fi
}

update_or_check_files_names_listing "$@"
```

2.32 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts

```

2.33 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

2.34 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

2.35 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

2.36 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

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-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
```

```
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
```

```
77  }
78  ol.level2{
79    counter-reset: level2;
80  }
81  ol.level1 li:before{
82    content: counter(level1) ". ";
83    counter-increment: level1;
84  }
85  ol.level2 li:before{
86    content: counter(level1) "." counter(level2) " ";
87    counter-increment: level2;
88  }
89  </style>
90  <title>@repository_name@</title>
91  </head>
92  <body >
93  <table class="title">
94    <tr>
95      <td style="padding:1ex">
96        <h1 class="titlemain">@repository_name@</h1>
97        <h2 class="titlerest">
98          @author_full_name@
99          <br>
100         <a href="@author_website@">
101           <span style="font-family:monospace">
102           @author_website@
103           </span>
104         </a>
105         <br>
106         <a href="mailto:@author_email@">
107           <span style="font-family:monospace">
108           @author_email@
109           </span>
110         </a>
111       </h2>
112     </td>
113   </tr>
114 </table>
115
116 <blockquote class="abstract">
117   <span style="font-weight:bold">Abstract:</span>
118   @abstract@
119 </blockquote>
120 <p>Current version: @current_date@</p>
121 <p>Current number of commits: @number_of_commits@</p>
122 <p>Current git SHA1: @current_git_SHA1@</p>
```

```
<p>
Code lines: @number_of_lines@
</p>


<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Listing of files</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
```

```
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, pip, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

<p>
Bash can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/bash/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/bash/</span></a>,<br>
<a href="https://tiswww.case.edu/php/chet/bash/bashtop.html"><span
```

```
  style="font-family:monospace"
>https://tiswww.case.edu/php/chet/bash/bashtop.html</span></a>,<br>
<a href="https://savannah.gnu.org/projects/bash/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/bash/</span></a>.
</p>

<p>
GNU coreutils can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/coreutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/coreutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/coreutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/coreutils/</span></a>.
</p>

<p>
uutils coreutils can be found at the following URLs:<br>
<a href="https://uutils.github.io/coreutils/"><span
  style="font-family:monospace"
>https://uutils.github.io/coreutils/</span></a>,<br>
<a href="https://github.com/uutils/coreutils"><span
  style="font-family:monospace"
>https://github.com/uutils/coreutils</span></a>.
</p>

<p>
diff (GNU diffutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/diffutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/diffutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/diffutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/diffutils/</span></a>.
</p>

<p>
find, xargs (GNU findutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/findutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/findutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/findutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/findutils/</span></a>.
</p>
```

```
<p>
grep (GNU grep) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/grep/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/grep/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/grep/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/grep/</span></a>.
</p>

<p>
sed (GNU sed) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/sed/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/sed/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/sed/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/sed/</span></a>.
</p>

<p>
ShellCheck can be found at the following URLs:<br>
<a href="https://www.shellcheck.net/"><span
  style="font-family:monospace"
>https://www.shellcheck.net/</span></a>,<br>
<a href="https://github.com/koalaman/shellcheck"><span
  style="font-family:monospace"
>https://github.com/koalaman/shellcheck</span></a>.
</p>

<p>
GNU Wget can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/wget/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/wget/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/wget/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/wget/</span></a>.
</p>

<p>
more of util-linux can be found at the following URLs:<br>
<a href="https://www.kernel.org/pub/linux/utils/util-linux/"><span
  style="font-family:monospace"
>https://www.kernel.org/pub/linux/utils/util-linux/</span></a>,<br>
```

```
<a
href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
><span
  style="font-family:monospace"
>https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
></a>,<br>
<a href="https://github.com/util-linux/util-linux"><span
  style="font-family:monospace"
>https://github.com/util-linux/util-linux</span></a> official?
</p>

<p>
pandoc can be found at the following URLs:<br>
<a href="https://pandoc.org/"><span
  style="font-family:monospace"
>https://pandoc.org/</span></a>,<br>
<a href="https://github.com/jgm/pandoc"><span
  style="font-family:monospace"
>https://github.com/jgm/pandoc</span></a>.
</p>

<p>
pdftex/pdflatex can be found at the following URLs:<br>
<a href="https://www.tug.org/applications/pdftex/"><span
  style="font-family:monospace"
>https://www.tug.org/applications/pdftex/</span></a>,<br>
<a href="https://tug.org/texlive/svn/"><span
  style="font-family:monospace"
>https://tug.org/texlive/svn/</span></a>.
</p>

<p>
tree can be found at the following URLs:<br>
<a href="https://oldmanprogrammer.net/source.php?dir=projects/tree"
><span style="font-family:monospace"
>https://oldmanprogrammer.net/source.php?dir=projects/tree</span></a>,<br>
<a href="https://gitlab.com/OldManProgrammer/unix-tree"><span
  style="font-family:monospace"
>https://gitlab.com/OldManProgrammer/unix-tree</span></a>,<br>
<a href="https://github.com/Old-Man-Programmer/tree"><span
  style="font-family:monospace"
>https://github.com/Old-Man-Programmer/tree</span></a>.
</p>

<p>
git can be found at the following URLs:<br>
```

```
<a href="https://git-scm.com/about"><span
  style="font-family:monospace"
>https://git-scm.com/about</span></a>,<br>
<a href="https://github.com/git/git"><span
  style="font-family:monospace"
>https://github.com/git/git</span></a>.
</p>

<p>
pylint can be found at the following URLs:<br>
<a href="https://www.pylint.org/"><span
  style="font-family:monospace"
>https://www.pylint.org/</span></a>,<br>
<a href="https://pylint.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://pylint.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/pylint-dev/pylint"><span
  style="font-family:monospace"
>https://github.com/pylint-dev/pylint</span></a>,<br>
<a href="https://pypi.org/project/pylint/"><span
  style="font-family:monospace"
>https://pypi.org/project/pylint/</span></a>.
</p>

<p>
notify-send can be found at the following URLs:<br>
<a href="https://gitlab.gnome.org/GNOME/libnotify/"><span
  style="font-family:monospace"
>https://gitlab.gnome.org/GNOME/libnotify/</span></a>,<br>
<a href="https://gnome.pages.gitlab.gnome.org/libnotify/"><span
  style="font-family:monospace"
>https://gnome.pages.gitlab.gnome.org/libnotify/</span></a>.
</p>

<p>
black can be found at the following URLs:<br>
<a href="https://github.com/psf/black"><span
  style="font-family:monospace"
>https://github.com/psf/black</span></a>,<br>
<a href="https://black.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://black.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://pypi.org/project/black/"><span
  style="font-family:monospace"
>https://pypi.org/project/black/</span></a>.
</p>
```

```
<p>
isort can be found at the following URLs:<br>
<a href="https://pycqa.github.io/isort/"><span
  style="font-family:monospace"
>https://pycqa.github.io/isort/</span></a>,<br>
<a href="https://github.com/pycqa/isort/"><span
  style="font-family:monospace"
>https://github.com/pycqa/isort/</span></a>,<br>
<a href="https://pypi.org/project/isort/"><span
  style="font-family:monospace"
>https://pypi.org/project/isort/</span></a>.
</p>

<p>
mypy can be found at the following URLs:<br>
<a href="https://mypy-lang.org/"><span
  style="font-family:monospace"
>https://mypy-lang.org/</span></a>,<br>
<a href="https://mypy.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://mypy.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/python/mypy"><span
  style="font-family:monospace"
>https://github.com/python/mypy</span></a>,<br>
<a href="https://pypi.org/project/mypy/"><span
  style="font-family:monospace"
>https://pypi.org/project/mypy/</span></a>.
</p>

<p>
pip can be found at the following URLs:<br>
<a href="https://pip.pypa.io/en/stable/"><span
  style="font-family:monospace"
>https://pip.pypa.io/en/stable/</span></a>,<br>
<a href="https://github.com/pypa/pip"><span
  style="font-family:monospace"
>https://github.com/pypa/pip</span></a>,<br>
<a href="https://pypi.org/project/pip/"><span
  style="font-family:monospace"
>https://pypi.org/project/pip/</span></a> (egg and chicken ;) ).
</p>

<p>
ruff can be found at the following URLs:<br>
<a href="https://docs.astral.sh/ruff/"><span
```

```
  style="font-family:monospace"
>https://docs.astral.sh/ruff/</span></a>,<br>
<a href="https://github.com/astral-sh/ruff"><span
  style="font-family:monospace"
>https://github.com/astral-sh/ruff</span></a>,<br>
<a href="https://pypi.org/project/ruff/"><span
  style="font-family:monospace"
>https://pypi.org/project/ruff/</span></a>.
</p>

<p>
bandit can be found at the following URLs:<br>
<a href="https://bandit.readthedocs.io/en/latest/"><span
  style="font-family:monospace"
>https://bandit.readthedocs.io/en/latest/</span></a>,<br>
<a href="https://github.com/PyCQA/bandit"><span
  style="font-family:monospace"
>https://github.com/PyCQA/bandit</span></a>,<br>
<a href="https://pypi.org/project/bandit/"><span
  style="font-family:monospace"
>https://pypi.org/project/bandit/</span></a>.
</p>

<p>
phpmd can be found at the following URLs:<br>
<a href="https://phpmd.org/"><span
  style="font-family:monospace"
>https://phpmd.org/</span></a>,<br>
<a href="https://github.com/phpmd/phpmd"><span
  style="font-family:monospace"
>https://github.com/phpmd/phpmd</span></a>,<br>
<a href="https://packagist.org/packages/phpmd/phpmd"><span
  style="font-family:monospace"
>https://packagist.org/packages/phpmd/phpmd</span></a>.
</p>

<p>
php can be found at the following URLs:<br>
<a href="https://www.php.net/"><span
  style="font-family:monospace"
>https://www.php.net/</span></a>,<br>
<a href="https://github.com/php/php-src"><span
  style="font-family:monospace"
>https://github.com/php/php-src</span></a>.
</p>

```

```
<p>
tex can be found at the following URL:<br>
<a href="https://ctan.org/tex-archive/systems/knuth/dist/tex"><span
  style="font-family:monospace"
>https://ctan.org/tex-archive/systems/knuth/dist/tex</span></a>.
</p>

<p>
pcregrep, pcre2grep can be found at the following URLs:<br>
<a href="https://www.pcre.org/"><span
  style="font-family:monospace"
>https://www.pcre.org/</span></a>,<br>
<a href="https://pcre2project.github.io/pcre2/"><span
  style="font-family:monospace"
>https://pcre2project.github.io/pcre2/</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre1"><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre1</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre2"><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre2</span></a>,<br>
<a
href="https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c</span
></a>.
</p>

<p>
python3 can be found at the following URLs:<br>
<a href="https://www.python.org/"><span
  style="font-family:monospace"
>https://www.python.org/</span></a>,<br>
<a href="https://docs.python.org/3/"><span
  style="font-family:monospace"
>https://docs.python.org/3/</span></a>.
</p>

<p>
If you see something that I missed regarding my dependencies,
please tell me/email me :).
</p>

@files_listing_HTMLPreInput@

```

```
<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

2.37 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
```

```
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
```

```
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Listing of files}
\label{section:listing}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
```

```
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, pip, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:\newline
\url{https://www.gnu.org/software/bash/},\newline
\url{https://tiswww.case.edu/php/chet/bash/bashtop.html},\newline
\url{https://savannah.gnu.org/projects/bash/}.

GNU coreutils can be found at the following URLs:\newline
\url{https://www.gnu.org/software/coreutils/},\newline
\url{https://savannah.gnu.org/projects/coreutils/}.

uutils coreutils can be found at the following URLs:\newline
\url{https://uutils.github.io/coreutils/},\newline
\url{https://github.com/uutils/coreutils}.

diff (GNU diffutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/diffutils/},\newline
\url{https://savannah.gnu.org/projects/diffutils/}.
```

```
find, xargs (GNU findutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/findutils/},\newline
\url{https://savannah.gnu.org/projects/findutils/}.

grep (GNU grep) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/grep/},\newline
\url{https://savannah.gnu.org/projects/grep/}.

sed (GNU sed) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/sed/},\newline
\url{https://savannah.gnu.org/projects/sed/}.

ShellCheck can be found at the following URLs:\newline
\url{https://www.shellcheck.net/},\newline
\url{https://github.com/koalaman/shellcheck}.

GNU Wget can be found at the following URLs:\newline
\url{https://www.gnu.org/software/wget/},\newline
\url{https://savannah.gnu.org/projects/wget/}.

more of util-linux can be found at the following URLs:\newline
\url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
},
\newline
\url{https://github.com/util-linux/util-linux} official?

pandoc can be found at the following URLs:\newline
\url{https://pandoc.org/},\newline
\url{https://github.com/jgm/pandoc}.

pdftex/pdflatex can be found at the following URLs:\newline
\url{https://www.tug.org/applications/pdftex/},\newline
\url{https://tug.org/texlive/svn/}.

tree can be found at the following URLs:\newline
\url{https://oldmanprogrammer.net/source.php?dir=projects/tree},\newline
\url{https://gitlab.com/OldManProgrammer/unix-tree},\newline
\url{https://github.com/Old-Man-Programmer/tree}.

git can be found at the following URLs:\newline
\url{https://git-scm.com/about},\newline
\url{https://github.com/git/git}.
```

```
pylint can be found at the following URLs:\newline
\url{https://www.pylint.org/},\newline
\url{https://pylint.readthedocs.io/en/stable/},\newline
\url{https://github.com/pylint-dev/pylint},\newline
\url{https://pypi.org/project/pylint/}.

notify-send can be found at the following URLs:\newline
\url{https://gitlab.gnome.org/GNOME/libnotify/},\newline
\url{https://gnome.pages.gitlab.gnome.org/libnotify/}.

black can be found at the following URLs:\newline
\url{https://github.com/psf/black},\newline
\url{https://black.readthedocs.io/en/stable/},\newline
\url{https://pypi.org/project/black/}.

isort can be found at the following URLs:\newline
\url{https://pycqa.github.io/isort/},\newline
\url{https://github.com/pycqa/isort/},\newline
\url{https://pypi.org/project/isort/}.

mypy can be found at the following URLs:\newline
\url{https://mypy-lang.org/},\newline
\url{https://mypy.readthedocs.io/en/stable/},\newline
\url{https://github.com/python/mypy},\newline
\url{https://pypi.org/project/mypy/}.

pip can be found at the following URLs:\newline
\url{https://pip.pypa.io/en/stable/},\newline
\url{https://github.com/pypa/pip},\newline
\url{https://pypi.org/project/pip/} (egg and chicken ;) ).

ruff can be found at the following URLs:\newline
\url{https://docs.astral.sh/ruff/},\newline
\url{https://github.com/astral-sh/ruff},\newline
\url{https://pypi.org/project/ruff/}.

bandit can be found at the following URLs:\newline
\url{https://bandit.readthedocs.io/en/latest/},\newline
\url{https://github.com/PyCQA/bandit},\newline
\url{https://pypi.org/project/bandit/}.

phpmd can be found at the following URLs:\newline
\url{https://phpmd.org/},\newline
\url{https://github.com/phpmd/phpmd},\newline
\url{https://packagist.org/packages/phpmd/phpmd}.

```

```
php can be found at the following URLs:\newline
\url{https://www.php.net/},\newline
\url{https://github.com/php/php-src}.

tex can be found at the following URL:\newline
\url{https://ctan.org/tex-archive/systems/knuth/dist/tex}.

pcregrep, pcre2grep can be found at the following URLs:\newline
\url{https://www.pcre.org/},\newline
\url{https://pcre2project.github.io/pcre2/},\newline
\url{https://github.com/PCRE2Project/pcre1},\newline
\url{https://github.com/PCRE2Project/pcre2},\newline
{\small
  \url{https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c}
}.

python3 can be found at the following URLs:\newline
\url{https://www.python.org/},\newline
\url{https://docs.python.org/3/}.

If you see something that I missed regarding my dependencies,
please tell me/email me :).

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

2.38 build_and_checks_variables/files_listing.tex.tpl

```
\subsection{
  build\_and\_checks\_dependencies/build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/build\_md\_from\_printable\_md.exec.sh
}
```

```
\label{
  build_and_checks_dependencies:build_md_from_printable_mdexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_beginnings.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_beginningslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_indentation.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_indentationlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/check\_URLs.libr.sh
}
\label{
  build_and_checks_dependencies:check_URLslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_URLs.libr.sh
}

```

```
\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/common\_options.libr.sh
}
\label{
  build_and_checks_dependencies:common_optionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_options.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/comparisons.libr.sh
}
\label{
  build_and_checks_dependencies:comparisonslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/comparisons.libr.sh
```

```
}


\subsection{
  build\_and\_checks\_dependencies/create\_PDF.exec.sh
}
\label{
  build_and_checks_dependencies:create_PDFexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/create_PDF.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/generate\_from\_template.libr.sh
}
\label{
  build_and_checks_dependencies:generate_from_templatelibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/generate_from_template.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
get\_common\_text\_glob\_patterns.libr.sh
}
\label{
  build_and_checks_dependencies:get_common_text_glob_patternslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/grammar\_and\_spell\_check.libr.sh
}
\label{
  build_and_checks_dependencies:grammar_and_spell_checklibrsh
}
```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
build\_licenses\_templates.exec.sh
}
\label{
  build_and_checks_dependencies:licenses_templates:%
build_licenses_templatesexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
build_licenses_templates.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_AGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_AGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_AGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_GPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_GPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
```

```
build_and_checks_dependencies/licenses_templates/%
license_file_header_GPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_LGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_LGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_LGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/lines\_counts.libr.sh
}
\label{
  build_and_checks_dependencies:lines_countslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_counts.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_filters.libr.sh
}
\label{
  build_and_checks_dependencies:lines_filterslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_filters.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_maps.libr.sh
```

```
}
\label{
  build_and_checks_dependencies:lines_mapslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_maps.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/overwrite\_if\_not\_equal.libr.sh
}
\label{
  build_and_checks_dependencies:overwrite_if_not_equallibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_black\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_black_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_black_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_isort\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_isort_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_isort_complement.libr.sh
}
```

```
\subsection{
  build\_and\_checks\_dependencies/repository\_data.libr.sh
}
\label{
  build_and_checks_dependencies:repository_datalibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/repository_data.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/shell\_checks\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:shell_checks_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/shell_checks_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.exec.php
}
\label{
  build_and_checks_dependencies:split_scoreexecphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.exec.php
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.libr.php
}
\label{
  build_and_checks_dependencies:split_scorelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.libr.php
}
```

```
\subsection{
  build\_and\_checks\_dependencies/strings\_functions.libr.sh
}
\label{
  build_and_checks_dependencies:strings_functionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/strings_functions.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/too\_long\_code\_lines.libr.sh
}
\label{
  build_and_checks_dependencies:too_long_code_lineslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/too_long_code_lines.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:update_common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_or\_check\_files\_names\_listing.exec.sh
}
\label{
  build_and_checks_dependencies:update_or_check_files_names_listingexecsh
}
```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh
}


\subsection{
  build\_and\_checks\_variables/.gitignore
}
\label{
  build_and_checks_variables:gitignore
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/.gitignore
}


\subsection{
  build\_and\_checks\_variables/bandit.ini
}
\label{
  build_and_checks_variables:banditini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit.ini
}


\subsection{
  build\_and\_checks\_variables/bandit\_baseline.json
}
\label{
  build_and_checks_variables:bandit_baselinejson
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit_baseline.json
}


\subsection{
  build\_and\_checks\_variables/black.toml
}
\label{
```

```
  build_and_checks_variables:blacktoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/black.toml
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.html.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptshtmltpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.tex.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptstextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_listing.tex.tpl
}
\label{
  build_and_checks_variables:files_listingtextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_listing.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_names\_listing.txt
```

```
}
\label{
  build_and_checks_variables:files_names_listingtxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_names_listing.txt
}


\subsection{
  build\_and\_checks\_variables/licenses\_notes.txt
}
\label{
  build_and_checks_variables:licenses_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/licenses_notes.txt
}


\subsection{
  build\_and\_checks\_variables/phpmd\_baseline.xml
}
\label{
  build_and_checks_variables:phpmd_baselinexml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/phpmd_baseline.xml
}


\subsection{
  build\_and\_checks\_variables/pylintrc.toml
}
\label{
  build_and_checks_variables:pylintrctoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/pylintrc.toml
}
```

```
\subsection{
  build\_and\_checks\_variables/repository\_data.txt
}
\label{
  build_and_checks_variables:repository_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/repository_data.txt
}


\subsection{
  build\_and\_checks\_variables/ruff.toml
}
\label{
  build_and_checks_variables:rufftoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/ruff.toml
}


\subsection{
  build\_and\_checks\_variables/shellcheck.ini
}
\label{
  build_and_checks_variables:shellcheckini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck.ini
}


\subsection{
  build\_and\_checks\_variables/shellcheck\_notes.txt
}
\label{
  build_and_checks_variables:shellcheck_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck_notes.txt
}
```

```
\subsection{
  build\_and\_checks.exec.sh
}
\label{
  build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks.exec.sh
}


\subsection{
  COPYING
}
\label{
  COPYING
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING
}


\subsection{
  COPYING.LESSER
}
\label{
  COPYINGLESSER
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING.LESSER
}


\subsection{
  git\_helpers/git\_checks.libr.sh
}
\label{
  git_helpers:git_checkslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
```

```
git_helpers/git_checks.libr.sh
}


\subsection{
  git\_helpers/post-commit
}
\label{
  git_helpers:post-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post-commit
}


\subsection{
  git\_helpers/post\_commit\_renaming\_comment.libr.sh
}
\label{
  git_helpers:post_commit_renaming_commentlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post_commit_renaming_comment.libr.sh
}


\subsection{
  git\_helpers/pre-commit
}
\label{
  git_helpers:pre-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/pre-commit
}


\subsection{
  IMAP\_transfert.py
}
\label{
  IMAP_transfertpy
}
```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
IMAP_transfert.py
}


\subsection{
  LiensUtiles.txt
}
\label{
  LiensUtilestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
LiensUtiles.txt
}


\subsection{
  miscellaneous/funny.sh
}
\label{
  miscellaneous:funnysh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/funny.sh
}


\subsection{
  miscellaneous/spampty\_data.txt
}
\label{
  miscellaneous:spampty_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/spampty_data.txt
}


\subsection{
  miscellaneous/%
yet\_another\_long\_long\_long\_long\_long\_long\_long\_long\_long\_filena%
me\_so\_that\_you\_can\_see\_that\_it\_works.txt
```

```
}
\label{
  miscellaneous:%
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_%
you_can_see_that_it_workstxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/%
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_%
you_can_see_that_it_works.txt
}


\subsection{
  README.md.tpl
}
\label{
  READMEmdtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
README.md.tpl
}


\subsection{
  unicode.libr.php
}
\label{
  unicodelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
unicode.libr.php
}


\subsection{
  upgrade\_build\_and\_checks.exec.sh
}
\label{
  upgrade_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
```

```
upgrade_build_and_checks.exec.sh
}


\subsection{
  wget\_sha512.libr.sh
}
\label{
  wget_sha512librsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
wget_sha512.libr.sh
}
```

2.39 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
```

```
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
```

```
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_listing.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456\
56
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456\
56\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456\
5\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123\
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
```

```
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

2.40 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.


Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
```

```
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
```

```
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )
```

```

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

2.41 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"/>
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/split_score.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
```

```
    file="build_and_checks_dependencies/split_score.libr.php"/>
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\CleanCode\MissingImport"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="unicode.libr.php"/>
</phpmd-baseline>
```

2.42 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

2.43 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

grammar_or_spell_checker_command=more


Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
```

```
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
        "0.1.0": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "4.4.4" : {}
          }
        }
      }
    }
  },
  "B": {
    "2.2.2": {
      "C": {
        // Dumb, you did not found it by yourself? Failed exam.
```

```
          "4.4.4" : {}
        }
      }
    }
}
Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.

bandit_venv=../venv_bandit
black_venv=../venv_black
isort_venv=../venv_isort
mypy_venv=../venv_mypy
pylint_venv=../venv_pylint
ruff_venv=../venv_ruff


upgrade venvs can be auto or ask, any other value is interpreted
as ask.

upgrade_venvs=ask


time interval can be 0 to always upgrade or ask to upgrade,
or a number at most bash max int
https://superuser.com/questions/1030122/
what-is-the-maximum-value-of-a-numeric-bash-shell-variable
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800
```

2.44 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

2.45 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"

```

2.46 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
           False positive when c contains \n.
- SC2034 : Unused variable
           False positive for globals that are declared in functions
           that are not currently called.
           Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
           Use printf '..%s..' "$foo".
           False positive when using a variable to build the format string.
- SC2086 : Double quote to prevent globbing and word splitting.
           False positive when returning integer variable
           via an indirection like return ${!foo}.
           See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
           False positive when passing arguments would yield
           a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
           False positives when some arguments are meant for debug in CLI.
```

```
- SC2125 : Brace expansions and globs are literal.
           False positive when assigning a variable declared with -i.
           https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
           False positive when using "$@" line return and pipes.
           https://github.com/koalaman/shellcheck/issues/3192
           Example:
             # shellcheck disable=SC2145
             all_self_code_lines "$@"\
               | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
           Very useful but false positive when I play with
           variable variable names.
- SC2162X: read without -r will mangle backslashes.
           False positive when read is used without variable:
           https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
           False positive on returning integer values.
           https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
             Use = to compare as string (or expand explicitly with $var).
           Very useful but false positive when I play with
           variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
           value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
           https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$" -e "=SC2120$"\
  -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$" -e "=SC2309$"\
  -e "=SC2312$" -e "=SC2154,SC2309$"
```

2.47 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    {
      printf "temp/\n\n"
    } > "${LFBFL_variables_directory}/.gitignore"
```

```
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='1d7cb16e64217df2171f5b70e6b4b65280ecc1953980c'
  LFBFL_correct_sha512+='376ccdcbf282b263b9e20fbcf46f3bb237a8f8255061'
  LFBFL_correct_sha512+='1079153989c29c5560929a8edc30c8fe6f516b4'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed_point_build* ]]; then
  printf -- "--fixed_point_build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
```

```
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed_point_build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.48 COPYING

```
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```
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    copy of the Corresponding Source for all the software in the
    product that is covered by this License, on a durable physical
```

```
    medium customarily used for software interchange, for a price no
    more than your reasonable cost of physically performing this
    conveying of source, or (2) access to copy the
    Corresponding Source from a network server at no charge.

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    only if you received the object code with such an offer, in accord
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```
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```

```
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   whatever part of its purpose remains meaningful, or

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```

```
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        manner specified by section 6 of the GNU GPL for conveying
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        Library.  A suitable mechanism is one that (a) uses at run time
        a copy of the Library already present on the user's computer
        system, and (b) will operate properly with a modified version
        of the Library that is interface-compatible with the Linked
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    e) Provide Installation Information, but only if you would otherwise
    be required to provide such information under section 6 of the
    GNU GPL, and only to the extent that such information is
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    Combined Work produced by recombining or relinking the
    Application with a modified version of the Linked Version. (If
    you use option 4d0, the Installation Information must accompany
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    a) Accompany the combined library with a copy of the same work based
```

```
    on the Library, uncombined with any other library facilities,
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```

2.50 git_helpers/git_checks.libr.sh

```
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# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle"
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits -> #Prefetch()"
  declare -i LFBFL_error=0
  git diff --cached --name-only\
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
    LFBFL_error=1
```

```
  fi
  if grep serializer /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
    LFBFL_error=1
  fi
  rm /tmp/DOSAS_django_git_check1.temp
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}


check_no_abusive_trailing_comma(){
  # Useful for Python code, although it regularly yields
  # false positives in Python code
  # (but it is easy to distinguish by hand).
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION"
  local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
  LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
  readonly LFBFL_send_body_1
  declare -i LFBFL_error=0
  git diff --cached -r\
    > /tmp/DOSAS_django_git_check2.temp
  if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    LFBFL_error=1
  fi
  rm /tmp/DOSAS_django_git_check2.temp
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_files_string=$(
    git diff --cached --name-only\
    | grep '\.py'
  )
  printf "%s\n" "${LFBFL_files_string}"
  declare -a LFBFL_some_files
  mapfile -t LFBFL_some_files <<< "${LFBFL_files_string}"
  readonly LFBFL_some_files
  declare -i LFBFL_error=0
```

```
  # Not very useful since LFBFL_file should be local to the loop by
  # default.
  local LFBFL_file
  for LFBFL_file in "${LFBFL_some_files[@]}"; do
    printf "Black will check formatting on file %s\n" "${LFBFL_file}"
    if ! black --check --diff "${LFBFL_file}";
    then
      LFBFL_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}
```

2.51 git_helpers/post-commit

```
#!/usr/bin/env bash
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#
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# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
```

```
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="post_commit_renaming_comment.libr.sh"
# shellcheck source=post_commit_renaming_comment.libr.sh
source "${LFBFL_library}"
LFBFL_library="../DevOrSysAdminScripts/build_and_checks_dependencies/"
LFBFL_library+="lines_filters.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "${LFBFL_library}"

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

2.52 git_helpers/post_commit_renaming_comment.libr.sh

```
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```

```
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#
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# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_cd_result2
pushd .
cd "${LFBFL_some_directory}" || {
  LFBFL_cd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no such directory"
  # shellcheck disable=SC2248
  exit ${LFBFL_cd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_popd_result2
popd || {
  LFBFL_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd"
  # shellcheck disable=SC2248
  exit ${LFBFL_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
```

```
66  # parametered with `git`. This point could be enhanced.
67  commit_a_file_renamed_comment(){
68    # Options:
69    #   --verbose
70    #   --log-directory-change
71    #   --max-comment-line-length
72    declare -i LFBFL_i_verbose=0
73    get_verbose_option "$@"
74    local LFBFL_work_directory=""
75    get_work_directory_option "$@"
76    local LFBFL_arg
77    local LFBFL_s_format
78
79    declare -i LFBFL_log_directory_change=0
80    if [[ "$*" == *--log-directory-change* ]]; then
81      LFBFL_log_directory_change=1
82    fi
83    readonly LFBFL_log_directory_change
84
85    declare -i LFBFL_max_comment_line_length=70
86    for LFBFL_arg in "$@"; do
87      if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
88        LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
89        LFBFL_max_comment_line_length=$((LFBFL_arg))
90        break
91      fi
92    done
93    readonly LFBFL_max_comment_line_length
94
95    if [[ LFBFL_i_verbose -eq 1 ]]; then
96      LFBFL_s_format="LFBFL_log_directory_change: %s\n"
97      LFBFL_s_format+="LFBFL_max_comment_line_length: %s\n"
98      # shellcheck disable=SC2059
99      printf "${LFBFL_s_format}"\
100       "${LFBFL_log_directory_change}"\
101       "${LFBFL_max_comment_line_length}"
102   fi
103   #-----------------------------------------------------------------------
104
105   enhanced_set_shell_option pipefail\
106     && trap 'enhanced_unset_shell_option pipefail' RETURN
107
108   declare -r LFBFL_renaming_lines=$(
109     git log -p -1\
110     | grep '^diff --git' -A 1
111   )
```

```
112     if [[ LFBFL_i_verbose -eq 1 ]]; then
113       printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
114     fi
115
116     declare -a LFBFL_renaming_lines_array
117     mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
118     readonly LFBFL_renaming_lines_array
119
120     declare -r LFBFL_timestamp=$(date --iso-8601="minute")
121     if [[ LFBFL_i_verbose -eq 1 ]]; then
122       printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
123     fi
124     get_split_score_simple 1 "${LFBFL_max_comment_line_length}" _
125     LFBFL_split_score_command1="${get_split_score_result}"
126     LFBFL_split_score_command_properties1="${get_split_score_result2}"
127     get_split_score_simple 1 "${LFBFL_max_comment_line_length}" /
128     LFBFL_split_score_command2="${get_split_score_result}"
129     LFBFL_split_score_command_properties2="${get_split_score_result2}"
130     declare -i LFBFL_renaming_happened=0
131     declare -i LFBFL_i
132     declare -i LFBFL_j
133     # declare -r LFBFL_special_file_name="files_names_listing.txt"
134     declare -r LFBFL_sfn="files_names_listing.txt"
135     local LFBFL_diff_line
136     local LFBFL_similarity_line
137     local LFBFL_old_file_path
138     local LFBFL_old_file_name
139     local LFBFL_old_file_directory
140     local LFBFL_new_file_path
141     local LFBFL_new_file_name
142     local LFBFL_new_file_directory
143     local LFBFL_useful_file_name
144     local LFBFL_extension
145     local LFBFL_comment_prefix
146     local LFBFL_comment_prefix2
147     local LFBFL_old_file_name2
148     local LFBFL_old_file_directory2
149     local LFBFL_new_file_name2
150     local LFBFL_new_file_directory2
151     local LFBFL_new_comment
152     declare -i LFBFL_copyright_line_number
153     declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
154     declare -i LFBFL_line_count
155     declare -i LFBFL_lines_after
156     local LFBFL_temp_file_path
157     for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
```

```
    LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
    LFBFL_i=$((LFBFL_i + 1))
    LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
    LFBFL_i=$((LFBFL_i + 1))
    # We need to increment 3 times since grep adds "--" lines between.
    LFBFL_i=$((LFBFL_i + 1))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
        "${LFBFL_diff_line}"\
        "${LFBFL_similarity_line}"
    fi
    if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
      continue
    fi

    # Extract old file name from diff line. --------------------------
    LFBFL_old_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 3
    )
    # It starts by "a/".
    LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
    LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
    LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
    # Extract new file name from diff line. --------------------------
    LFBFL_new_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 4
    )
    # It starts by "b/".
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      LFBFL_s_format="LFBFL_old_file_path: %s\n"
      LFBFL_s_format+="LFBFL_old_file_name: %s\n"
      LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
      LFBFL_s_format+="LFBFL_new_file_path: %s\n"
      LFBFL_s_format+="LFBFL_new_file_name: %s\n"
      LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_old_file_path}"\
        "${LFBFL_old_file_name}"\
        "${LFBFL_old_file_directory}"\
        "${LFBFL_new_file_path}"\
```

```
        "${LFBFL_new_file_name}"\
        "${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. -----------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
        "${LFBFL_useful_file_name}"\
        "${LFBFL_extension}"
    fi
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
          "${LFBFL_useful_file_name}"\
          "${LFBFL_extension}"
      fi
    done
    LFBFL_comment_prefix=""
    if [[ "${LFBFL_extension}" == "sh" ]]; then
      LFBFL_comment_prefix="# "
    fi
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
    fi
    #-----------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"’"’
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
```

```
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_split_score_command_properties1}"
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_split_score_command_properties2}"
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_split_score_command_properties1}"
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_split_score_command_properties2}"
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. -----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
    then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_log_directory_change -eq 1 ]] && [[\
      "${LFBFL_new_file_directory}"\
```

```
296         != "${LFBFL_old_file_directory}"\
297       ]]; then
298         if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
299         then
300           LFBFL_new_comment+=$'\n'
301         fi
302         LFBFL_new_comment+="${LFBFL_comment_prefix}"
303         LFBFL_new_comment+="${LFBFL_timestamp}"
304         LFBFL_new_comment+=": This file was moved from"
305         LFBFL_new_comment+=$'\n'
306         LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
307         LFBFL_new_comment+=$'\n'
308         LFBFL_new_comment+="${LFBFL_comment_prefix}to"
309         LFBFL_new_comment+=$'\n'
310         LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
311       fi
312       if [[ LFBFL_i_verbose -eq 1 ]]; then
313         printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
314       fi
315       # Find line with ©Copyright. -------------------------------------
316       LFBFL_copyright_line_number=$(
317         grep -n '©Copyright' "${LFBFL_new_file_path}"\
318         | head --lines=1\
319         | cut -d ':' -f 1
320       )
321       if [[ LFBFL_i_verbose -eq 1 ]]; then
322         printf "LFBFL_copyright_line_number: %s\n"\
323           "${LFBFL_copyright_line_number}"
324       fi
325       if [[ LFBFL_copyright_line_number -eq 0 ]]; then
326         notify-send "${LFBFL_send_summary_2}"\
327           "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
328         if [[ LFBFL_i_verbose -eq 1 ]]; then
329           printf "No copyright line. Skipping file.\n"
330         fi
331         continue
332       fi
333       # Get total number of lines. -------------------------------------
334       LFBFL_line_count=$(
335         wc -l\
336         < "${LFBFL_new_file_path}"
337       )
338       if [[ LFBFL_i_verbose -eq 1 ]]; then
339         printf "LFBFL_line_count: %s\n" "${LFBFL_line_count}"
340       fi
341       # Compute number of lines after. ---------------------------------
```

```
    LFBFL_lines_after=$((
      LFBFL_line_count - LFBFL_copyright_line_number
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_lines_after: %s\n" "${LFBFL_lines_after}"
    fi
    # Update file. ---------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_copyright_line_number}"\
        "${LFBFL_new_file_path}"
      printf "%s\n" "${LFBFL_new_comment}"
      tail --lines="${LFBFL_lines_after}" "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. ---------------------------------------------
    git add "${LFBFL_new_file_path}"
    LFBFL_renaming_happened=1
  done
  readonly LFBFL_renaming_happened
  if [[ LFBFL_renaming_happened -eq 1 ]]; then
    # Commit ---------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    git commit --message="${LFBFL_message}"
  fi
}
```

2.53 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```
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#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="git_checks.libr.sh"
# shellcheck source=git_checks.libr.sh
source "${LFBFL_library}"

declare -ig LFBFL_git_hook_error=0
if ! check_files; then
  LFBFL_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_git_hook_error=1
fi
readonly LFBFL_git_hook_error
if [[ LFBFL_git_hook_error -eq 1 ]]; then
  exit 1
fi
exit 0
```

2.54 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

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```

```
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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


```

```
def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
```

```
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
```

```
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
    )
    parser.add_argument(
        "from_password",
        type=str,
        help="Email account password of user one.",
    )
    parser.add_argument(
        "to_login",
        type=str,
        help="Email account login of user two.",
    )
```

```
    parser.add_argument(
        "to_password",
        type=str,
        help="Email account password of user two.",
    )
    parser.add_argument(
        "--to_new_folder",
        type=str,
        default="",
        help="Transfert emails into a new folder of this name.",
    )
    parser.add_argument(
        "--from_host",
        type=str,
        default="imap.gmail.com",
        help="Host of first email account",
    )
    parser.add_argument(
        "--from_port",
        type=int,
        default=993,
        help="Port on host of first email account",
    )
    parser.add_argument(
        "--to_host",
        type=str,
        default="imap.gmail.com",
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
```

```
            "-----------------------------------------------------\n"
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

2.55 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
-----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
```

```
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
```

2.56 miscellaneous/funny.sh

```
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unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
```

```
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

2.57 miscellaneous/spampty_data.txt

```



a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

2.58 miscellaneous/yet_another_long_long_long_long_long_long_long_long_long_filename_s

2.59 README.md.tpl

```
# DevOrSysAdminScripts

[![CodeFactor-badge]][CodeFactor-package-page]
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[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |
```

```
Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.

But the files "pre-commit" and "post-commit" were not renamed,
since it is not possible to give them other names to use them as
pre-commit/post-commit hooks in `git`.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge
```

```
[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

[Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

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languages/top/llyaudet/DevOrSysAdminScripts

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languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json
```

```
[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

2.60 unicode.libr.php

```
<?php
/*
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This file was renamed from "unicode.php" to "unicode.libr.php".
*/

function decimal_code_point_to_UTF8(
  $i_code_point_in_decimal_notation
){
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
  if($i_code_point_in_decimal_notation < 128){
    return chr($i_code_point_in_decimal_notation);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 256 * (2**3)){
    $i_first_byte_base_value = 192;
    $i_last_byte_significant_bits = (
      $i_code_point_in_decimal_notation % (2**6)
    );
```

```
43      $i_first_byte_significant_bits = intdiv(
44        (
45          $i_code_point_in_decimal_notation
46          - $i_last_byte_significant_bits
47        ),
48        2**6,
49      );
50      $s_result = chr(
51        $i_first_byte_base_value
52        + $i_first_byte_significant_bits
53      );
54      $s_result .= chr(
55        $i_continuation_base_value
56        + $i_last_byte_significant_bits
57      );
58      return $s_result;
59    }
60    // 1110xxxx 10xxxxxx 10xxxxxx
61    if($i_code_point_in_decimal_notation < 256 * 256){
62      $i_first_byte_base_value = 224;
63      $arr_arr_data_per_byte_reverse = [
64        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
65        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
66        ["i_bits" => 4, "i_base_value" => $i_first_byte_base_value],
67      ];
68      $i_rest = $i_code_point_in_decimal_notation;
69      $s_result = "";
70      foreach($arr_arr_data_per_byte_reverse as $arr_data){
71        // _Pragma unroll
72        // J'aurais pu "roller" les 4 cas des ifs.
73        // Mais pour les perfs, il faudrait tout unroller vu le nombre
74        // d'itérations. Et j'aimerais bien comparer du code unrollé
75        // aux 2 niveaux à la main avec celui unrollé par le compilateur
76        // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
77        // l'intérieur d'abord (pour ne pas refaire le même boulot)
78        // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
79        // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
80        $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
81        $i_rest = intdiv(
82          $i_rest - $i_significant_bits,
83          2**$arr_data["i_bits"],
84        );
85        $s_result = (  // .= or =.
86          chr($i_significant_bits + $arr_data["i_base_value"])
87          .$s_result
88        );
```

```
    }
    return $s_result;
  }
  // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 256 * 256 * (2**5)){
    $i_first_byte_base_value = 240;
    $arr_arr_data_per_byte_reverse = [
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 3, "i_base_value" => $i_first_byte_base_value],
    ];
    $i_rest = $i_code_point_in_decimal_notation;
    $s_result = "";
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data["i_bits"],
      );
      $s_result = (  // .= or =.
        chr($i_significant_bits + $arr_data["i_base_value"])
        .$s_result
      );
    }
    return $s_result;
  }
  throw new Exception(
    "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
    ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
    ." 3 octets au lieu de 4 :face-palm:."
    ." Du coup, ça peut donner quelques \"surprises\"."
    ." Sinon, y'a aussi ASCII dans MySQL."
    ." Et comme ça joue sur la mémoire et les performances,"
    ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
    ." :)"
  );
}



function hexa_code_point_to_UTF8(
  $s_code_point_in_hexadecimal_notation
){
  $s_code_point_in_hexadecimal_notation = strtoupper(
```

```
135       $s_code_point_in_hexadecimal_notation
136     );
137     $i_code_point_in_decimal_notation = 0;
138     $i_ord_0 = ord('0');
139     $i_ord_A = ord('A');
140     $i_digit_offset = -$i_ord_0;
141     $i_letter_offset = 10 - $i_ord_A;
142     for(
143       $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
144       $i < $i_max;
145       ++$i
146     ){
147       $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
148       $i_code_point_in_decimal_notation *= 16;
149       if(ctype_digit($s_hexa_digit)){
150         $i_code_point_in_decimal_notation += (
151           ord($s_hexa_digit) + $i_digit_offset
152         );
153         continue;
154       }
155       if(ctype_xdigit($s_hexa_digit)){
156         $i_code_point_in_decimal_notation += (
157           ord($s_hexa_digit) + $i_letter_offset
158         );
159         continue;
160       }
161       throw new Exception("Not an hexadecimal digit.");
162     }
163     return decimal_code_point_to_UTF8(
164       $i_code_point_in_decimal_notation
165     );
166   }
167
168
169
170   function check_string_is_valid_ASCII($s_string){
171     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
172       if(ord($s_string[$i]) > 127){
173         throw new Exception(
174           "Non-ASCII character found at octet position "
175           .$i
176           ., value "
177           .ord($s_string[$i])
178           ."."
179         );
180       }
```

```
181     }
182     return true;
183   }
184
185
186
187   function check_file_is_valid_ASCII($s_file_name){
188     $s_string = file_get_contents($s_file_name);
189     if($s_string === false){
190       throw new Exception("File ".$s_file_name." not found.");
191     }
192     return check_string_is_valid_ASCII($s_string);
193   }
194
195
196
197   function check_string_is_valid_UTF8($s_string){
198     $i_continuation_octet_needed = 0;
199     $i_character_start_position = 0;
200     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
201       $i_current_octet = ord($s_string[$i]);
202       if($i_continuation_octet_needed > 0){
203         if($i_current_octet < 128 || $i_current_octet >= 192){
204           throw new Exception(
205             "Non-UTF8 character found at start position "
206             .$i_character_start_position
207             ., octet at position "
208             .$i
209             ." has value "
210             .$i_current_octet
211             ." which is not a continuation octet."
212           );
213         }
214         --$i_continuation_octet_needed;
215         continue;
216       }
217
218       $i_character_start_position = $i;
219       if($i_current_octet < 128){ // 0xxxxxxx ASCII
220         continue;
221       }
222       if($i_current_octet >= 128 && $i_current_octet < 192){
223         throw new Exception(
224           "Non-UTF8 character found at start position "
225           .$i_character_start_position
226           ., octet at position "
```

```
227          .$i
228          ." has value "
229          .$i_current_octet
230          ." which is a continuation octet."
231        );
232      }
233      if($i_current_octet >= 192 && $i_current_octet < 224){
234        // 110xxxxx 10xxxxxx
235        $i_continuation_octet_needed = 1;
236        continue;
237      }
238      if($i_current_octet >= 224 && $i_current_octet < 240){
239        // 1110xxxx 10xxxxxx 10xxxxxx
240        $i_continuation_octet_needed = 2;
241        continue;
242      }
243      if($i_current_octet >= 240 && $i_current_octet < 248){
244        // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
245        $i_continuation_octet_needed = 3;
246        continue;
247      }
248      throw new Exception(
249        "Non-UTF8 character found at start position "
250        .$i_character_start_position
251        ., octet at position "
252        .$i
253        ." has value "
254        .$i_current_octet
255        ." which is invalid."
256      );
257    }
258    return true;
259  }
260
261
262
263  function check_file_is_valid_UTF8($s_file_name){
264    $s_string = file_get_contents($s_file_name);
265    if($s_string === false){
266      throw new Exception("File ".$s_file_name." not found.");
267    }
268    return check_string_is_valid_UTF8($s_string);
269  }
270
271
272
```

```
/*
?>
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}
//*/

```

2.61 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
```

```
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# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed_point_build* ]]; then
    LFBFL_fixed_point_build="--fixed_point_build"
  fi
  readonly LFBFL_verbose
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  ./build_and_checks.exec.sh "." "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"
}

upgrade_build_and_checks "$@"
```

2.62 wget_sha512.libr.sh

```
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# This file is part of DevOrSysAdminScripts library.
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#
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# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1 file_path
  # $2 download_URL
  # $3 correct_sha512
  # $4 optional --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  if [[ ! -f "$1" ]]; then
    wget "${LFBFL_verbose}" -O "$1" "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum "$1"\
    | cut -f1 -d' '
  )
```

```
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
    return 1
  fi
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !